



**Modern University for Technology &
Information**



Faculty of Computers & Artificial Intelligence

Face Aging Using Generative Adversarial Networks (GANs)

Computer Science Graduation Project

Submitted By

Noha Atef Mohamed
Mona Ghareep Ahmed
Mohamed Magdy El-Komy
Omar Mohamed Talaat
Mohamed Mohsen ElDeeb
Islam Ahmed Sobh
Mohamed Magdy Mokhtar
Ahmed Magdy Mokhtar

Supervised By

DR. Marwa Refaie
T.A. Dina Ahmed
T.A. Abdelrahim Alnageh

Abstract

Face Aging refers to processing an image of a face to make it look older or younger. As the field of artificial intelligence continues to advance, the ability to accurately predict and model the aging process of human faces has become an increasingly important area of research. In this project, we propose a novel approach to face aging recognition using Generative Adversarial Networks (GANs).

GANs have shown remarkable success in generating realistic images by learning the underlying distribution of a dataset and synthesizing new samples that closely resemble the original data. Our project aims to leverage the power of GANs to create a model capable of accurately predicting the aging progression of facial images.

In this work, we needed to study this problem by using Generative Models and, more specifically, Generative Adversarial Networks. Our work relies on extensive research, reading previous experiences, and comparing them. Traditional methods often struggle with the intricate transformations faces undergo over time, GANs have emerged as a powerful tool for generating realistic images, making them ideal for this task. The aim is to implement and train some chosen methods on real data, propose an improvement to the models, and analyze the results to propose a face-aging solution. We have tried multiple hybrid models to achieve the best possible accuracy.

Acknowledgment

First and foremost, we are deeply thankful to ALLAH for granting us the strength and determination to see this project through to completion.

We would like to extend our appreciation to the Modern University for Technology and Information (MTI) for providing an environment that truly supports academic and research endeavors.

We would like to give special thanks to our amazing supervisor, **Dr. Marwa N. Refaie**. Your unwavering support, patience, and willingness to answer all our questions were crucial in shaping the direction and success of this research. We especially appreciate the emotional support you provided, which had a huge impact on our progress.

We also want to thank **Prof. Dr. Hesham El-Deeb**, head of the Artificial Intelligence Department, for his insightful guidance and support throughout this year.

A special thanks to **Prof. Dr. Hanafy Ismail**, head of the Computer Science Department, and **Prof. Dr. Mohammad Anwar Assal**, head of the IT Center at MTI University. Your leadership and support have been invaluable.

We are grateful to **Prof. Dr. Hafez Abd El-Wahab**, head of the Information Systems Department, as well as **Prof. Dr. Alaa Abd El-Raheem**, **Prof. Dr. Sayed Bakar**, and **Dr. Ahmed Saied** for their dedication and support throughout our studies.

Finally, a big thank you to **T.A. Dina Ahmed** and **T.A. Abdelrahim Alnageh**. Your continuous support and efforts to help us in every way possible have not gone unnoticed, and we are truly grateful.

Table of Contents

1.1 Introduction	2
1.2 Face Recognition History.....	3
1.3 Face Aging Overview.....	5
1.4 Methodology	6
1.4.1 Generative Adversarial Networks (GANs).....	6
1.4.2 CNN (Convolutional Neural Network) Model	20
1.5 Project Scope.....	21
1.5.1 Description of the Task	22
1.5.2 Problem Definition.....	23
1.5.3 Motivation	24
2.1 Introduction	26
2.2 The Planning Process	26
2.3 Project Tasks	29
2.3.1 TASK ASSIGNMENT.....	29
2.3.2 GANTT CHART	30
2.3.3 NETWORK DIAGRAM.....	31
3.1 Artificial Intelligence Overview	36
3.2 Face Recognition Concepts.....	38
3.2.1 Face Recognition Block Diagram	39
3.2.2 Pre-processing Faces.....	42
3.3 Age-Invariant Face Recognition	45
3.3.1 Age-Invariant Face Recognition	46
3.4 Related Work.....	47

3.4.2 Main Points and conclusions of Each Related Paper.....	50
4.1 GANs for Aging	70
4.1.1 Datasets	70
4.1.2 Generator Network Architecture.....	71
4.1.3 Discriminator Network Architecture.....	72
4.2 Face Verification and Recognition.....	73
4.3 User Interface with Tkinter	74
4.4 Conclusion.....	80
5.1 Generative adversarial Networks(GANs)	85
5.2 Face Recognition.....	88
5.2.1 Face Recognition without aging:	89
5.2.2 Age Invariant Face Recognition.....	90
6.1 Conclusion.....	93
6.2 Future Work	93
References	95

List of figures

Figure 1. Flow Of Recognition [22]	3
Figure 2. Primary stages in the history of face recognition [1].	4
Figure 3. Saroo Brierle a before he went missing at age 5 and b after reuniting with his biological mother at age 30 [2]	5
Figure 4. Face Aging effects across different ages of the same person [3]	6
Figure 5. How do GANs work? [27].....	7
Figure 6 . GANs Use Case examples.....	11
Figure 7. General Architecture of GANs [28]	15
Figure 8. A standard GAN architecture and loss function [5]..	16
Figure 9. GANs Vs CGANs [14].	18
Figure 10. Deep Convolutional GAN Architecture [15].	18
Figure 11 cycleGAN operates under the assumption that you have unpaired datasets from two domains (e.g., horse images and zebra images). The goal is to	20
Figure 12. General Architecture of CNN	21
Figure 13. New Facial Recognition system by American Airlines and Thales [23].....	22
Figure 14 Planning Process Cycle [11]	27
Figure 15 Task Assignment from the Allocation	30
Figure 16 Gantt Chart of Project Planning.....	31
Figure 17 Network Diagram of project planning.....	33
Figure 18 Network Diagram of project planning.....	
Figure 19 Network Diagram of project planning.....	
Figure 20 Draw.io	34
Figure 21 MS Project	34
Figure 22.....	Error! Bookmark not defined.
Figure 23 AI main fields are depicted in Figure12. Machine learning is the heart of AI [12].....	36
Figure 24. AI main fields	37
Figure 25 . Face-recognition system schema.....	38
Figure 26 Block diagram of the human face recognition system [6]	39
Figure 27 A complete structure of face recognition system. [26].....	42
Figure 28 Preprocessing for faces [13]	44
Figure 29 Demonstration of our aging simulation results (images in the first column are input faces of two subjects) [14]	45
Figure 30 Age-Invariant Example "FG-NET Data Set"	46
Figure 31. ML and DL Models	59
Figure 32. Decision Tree Algorithm in Machine Learning	60
Figure 33. Basic Concept of Decision Tree	Error! Bookmark not defined.

Figure 34. Support Vector Machine (SVM)	62
Figure 35.PCA (Principal Component Analysis).....	65
Figure 36.DWT processing generating two and three levels of resolution[11].	66
Figure 37 AGFW young faces(30-34 years old)	71
Figure 38 AGFW old faces (60-94 years old).....	71
Figure 39 Passport is valid and not expired	76
Figure 40 Passport is expired.....	76
Figure 41 GUI Workflow	79
Figure 42 A to B generated plot_004500	Error! Bookmark not defined.
Figure 43 Results during the training of the GAN model at iterations 3900, 4200, 4500	87
Figure 44 Results during the training of the GAN model at iterations 6900, 7200, 7500	87
Figure 45 Results during the second stage of training of the GAN model at iterations 2400, 3000, 3600	88

List of Abbreviations

FR	Face Recognition
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep learning
PCA	Principal Component Analysis
GAN	Generative Adversarial Networks
CNN	Convolutional Neural Network
SVM	Support Vector Machine
GUI	Graphical User Interface
CPU	Central Processing Unit
GPU	Graphics Processing Unit
API	Aplication Programming Interface

Chapter 1

Introduction

1.1 Introduction

In recent years, the field of artificial intelligence (AI) has witnessed remarkable advancements. One area of growing interest and significance is face aging recognition, which involves predicting the progression of facial appearance over time. This capability has numerous applications ranging from forensic age progression to cosmetic surgery simulation and biometric authentication systems. With the advent of machine learning and deep learning techniques, researchers have made significant strides in developing accurate and robust models for face aging prediction. In this documentation, we present our exploration into the realm of face aging recognition using machine learning and deep learning approaches, with a focus on Generative Adversarial Networks (GANs).

Face aging is an interesting and challenging problem to test the capabilities of Neural Networks. Also, it has become a widely studied problem because of the applications it offers. It can be used to predict the looks of missing people to make finding them easier, a very interesting tool for the police. Related to this, it could help with face recognition and face verification with age gaps, for instance when the available data is outdated. To address this complex problem, we are going to first look at what other researchers have experimented with and the results they have found to see what are the state-of-the-art methods.

1.2 Face Recognition History

Face recognition refers to the technology or process of identifying or verifying a person's identity based on their facial features. It involves capturing an image or video of a person's face and then analyzing it to determine whether it matches any known faces in a database or to verify the identity of the individual.

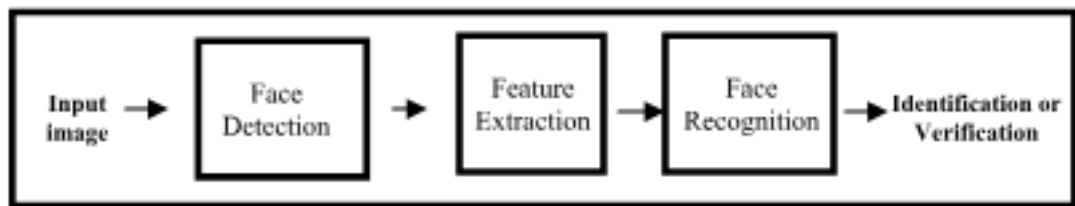


Figure 1. Flow Of Recognition [22]

This section reviews the most significant historical stages that have contributed to the advancement of face recognition technology [1]:

- **1964:** The American researchers studied facial recognition computer programming. They imagine a semi-automatic method, where operators are asked to enter twenty computer measures, such as the size of the mouth or the eyes.
- **1977:** The system was improved by adding 21 additional markers (e.g., lip width, hair color).
- **1988:** Artificial intelligence was introduced to develop previously used theoretical tools, which showed many weaknesses.
- **1991:** Alex Pentland and Matthew Turk of the Massachusetts Institute of Technology (MIT) presented the first successful example of facial recognition technology, which uses the statistical Principal component analysis (PCA) method.

- **1998:** the Defense Advanced Research Projects Agency (DARPA) developed the Face Recognition Technology (FERET) program, which provided the world with a sizable, challenging database composed of 2400 images for 850 persons.
- **2005:** The Face Recognition Grand Challenge (FRGC) competition was launched to encourage and develop face recognition technology designed to support existing facial recognition initiatives.
- **2011:** Everything accelerates due to deep learning, a machine learning method based on artificial neural networks. The computer selects the points to be compared: it learns better when it supplies more images.
- **2014:** Facebook knows how to recognize faces due to its internal algorithm, Deepface. The social network claims that its method approaches the performance of the human eye near to 97%.

Outlining of the most significant historical stages that have contributed to the advancement of face recognition technology

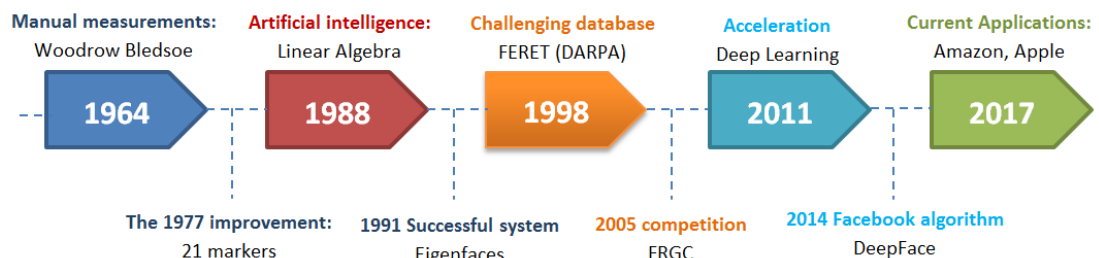


Figure 2. Primary stages in the history of face recognition [1].

1.3 Face Aging Overview

Face age progression, also known as face aging, predicts future appearance. At the same time, face age regression, also known as de-aging or rejuvenation, is an estimation of previous looks.

Recently, it has become a popular topic in computer vision, multimedia, and biometrics. Its real-world applications include finding missing children, digital entertainment, cosmetic surgery, cross-age face recognition, et [2]. The face is usually the only biometrics available to trace missing children, as parents and relatives are more likely to have images of the child, rather than other biometrics, such as fingerprints or irises. Saroo Brierley (also known as Sheru), was lost at the age of 5 and then reunited with his biological mother at age 30 years via Google Earth



(a)



(b)

Figure 3. Saroo Brierley a before he went missing at age 5 and b after reuniting with his biological mother at age 30 [2]

A lot of recent work focuses on predicting the age of a given image, to extract the age-determining features in a face image. the problem changes a lot when the output needs to be an altered image instead of a label: we are moving from Discriminative Models to Generative Models. One of the core goals when aging a face is to preserve the individual's identity. Hence, the resulting image needs to relate strongly to the input image.

Another core goal is to produce a visually convincing age effect on the image, avoiding blurry or non-realistic results. Furthermore, both goals need to be achieved with consistency: age(k) and rejuvenation (k) should attempt to be the inverse of one another [3].



Figure 4. Face Aging effects across different ages of the same person [3]

1.4 Methodology

1.4.1 Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs) represent a significant advancement in the field of artificial intelligence, particularly in the domain of generative models. Introduced by Ian Goodfellow [4] and his colleagues in 2014, GANs have revolutionized how machines can learn to create new data that mirrors real-world data. This breakthrough has found applications across various industries, transforming the way we approach problems in image processing, audio synthesis, data augmentation, and more.

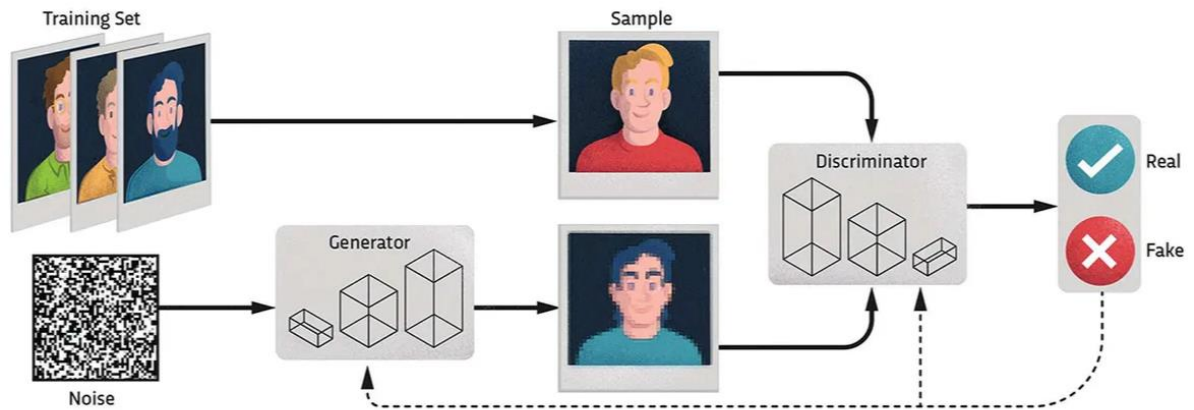


Figure 5. How do GANs work? [27]

Core Architecture of GANs

At the heart of GANs lies a unique architecture comprising two neural networks: the generator and the discriminator. These two networks are engaged in a continuous and dynamic adversarial process:

- **Generator:** The generator's role is to produce synthetic data that is as realistic as possible. It starts with random noise and uses it to generate data samples that mimic the characteristics of the real data set. Through backpropagation and gradient descent, the generator updates its parameters to create increasingly plausible data.
- **Discriminator:** The discriminator, on the other hand, evaluates the authenticity of the data. It is trained to distinguish between real data samples and those generated by the generator. The discriminator outputs a probability indicating whether a given input is real or fake.

The interaction between these two networks is formulated as a minimax game. The generator aims to maximize the probability of the discriminator

making a mistake, while the discriminator strives to minimize this probability. Over numerous iterations, this adversarial training results in the generator producing highly realistic data samples, as it learns to fool the discriminator more effectively [4].

Applications of GANs

- **Image Generation and Enhancement**

One of the most prominent applications of GANs is in the field of image generation and enhancement. GANs have demonstrated remarkable success in generating high-resolution, realistic images from scratch. They are also used for tasks such as: **Super-Resolution:** Enhancing the resolution of images, making them clearer and more detailed. For example, the Super-Resolution GAN (SRGAN) model can upscale images to higher resolutions while preserving details and textures.

- **Image Restoration:** GANs can restore old, damaged, or low-quality images. This includes removing noise, filling in missing parts, and correcting distortions. This application is particularly valuable in fields like digital archiving and historical preservation.
- **Style Transfer and Synthesis:** StyleGAN, developed by Nvidia, can generate images of human faces with such high realism that they are indistinguishable from actual photographs. This model allows for the manipulation of

various style parameters, enabling the creation of new and unique images.

- **Data Augmentation**

In many machine learning tasks, the availability of large and diverse datasets is crucial for training robust models.

However, acquiring sufficient data can be challenging, especially in specialized fields such as medical imaging.

GANs address this issue by generating synthetic data that can be used to augment existing datasets. This process improves the performance and generalization of machine learning models by providing them with more varied training examples.

- **Medical Imaging:**

GANs are used to generate additional medical images, such as MRI or CT scans, to enhance the training of diagnostic models. For instance, Frid-Adar et al. (2018) demonstrated the use of GANs to augment liver lesion datasets, leading to improved classification performance

- **Text-to-Image Synthesis**

Another fascinating application of GANs is text-to-image synthesis, where GANs generate images based on textual descriptions. This capability opens up new possibilities in areas such as:

- **Creative Design and Art:**

Artists and designers can use GANs to create images that match specific textual concepts, leading to innovative and unique artworks. Projects like DALL-E by OpenAI exemplify

this by generating diverse and imaginative images from detailed text prompts.

- **Virtual and Augmented Reality:**

GANs can create virtual environments and objects that align with narrative descriptions, enhancing the realism and immersion of virtual and augmented reality experiences.

- **Video Generation and Editing**

GANs are also making strides in the realm of video generation and editing. They can create realistic video sequences, predict future frames in a video, and enhance the resolution of video content. These capabilities are valuable for:

- **Content Creation:** GANs enable the creation of entirely synthetic videos for entertainment, educational content, and advertising. This technology can reduce production costs and time while allowing for creative storytelling.
- **Video Restoration:** Similar to image restoration, GANs can enhance the quality of old or degraded video footage, making it suitable for modern viewing standards. This application is useful for preserving historical recordings and improving the quality of archival footage.

- **Speech and Audio Synthesis**

In the domain of audio, GANs contribute to the generation of realistic speech and other sounds. These capabilities have significant implications for various applications: Virtual Assistants and Conversational Agents: GANs can generate natural-sounding speech for virtual assistants, making

interactions more lifelike and engaging. This enhances user experience in applications like smart home devices and customer service bots.



Figure 6 . GANs Use Case examples

- **Music Production:** Musicians and producers use GANs to generate new music tracks, explore different musical styles, and enhance audio quality. GANs can also be used to create backing tracks, sound effects, and other audio elements for multimedia projects.

Advantages of Using GANs

- **Realism and Quality**
One of the key strengths of GANs is their ability to produce highly realistic outputs. The adversarial training process pushes the generator to create data that is virtually indistinguishable from real data, resulting in high-quality synthetic samples. This realism is particularly important in applications where visual or auditory fidelity is crucial, such

as image and video generation, medical imaging, and audio synthesis.

- **Versatility**

GANs are versatile models that can be applied to various types of data, including images, audio, text, and video. This flexibility allows them to be used across a wide range of industries and applications. Whether it's generating realistic human faces, synthesizing new music tracks, or creating immersive virtual environments, GANs provide a powerful tool for diverse tasks.

- **Innovation in Creative Fields**

GANs have opened up new avenues for creativity and artistic expression. Artists, designers, and musicians use GANs to explore new creative possibilities, generating unique and novel content that was previously unattainable. This innovation is driving advancements in fields such as digital art, fashion design, music production, and more.

- **Improvement of Machine Learning Models**

By generating additional training data, GANs help improve the performance of machine learning models. This is particularly beneficial in scenarios where acquiring real data is challenging, expensive, or time-consuming. Synthetic data generated by GANs can augment existing datasets, providing more diverse and representative examples for training. This leads to better generalization and robustness in machine learning models.

Challenges and Future Directions

Despite their remarkable capabilities, GANs also present certain challenges and limitations. Training GANs can be difficult due to issues such as mode collapse, where the generator produces a limited variety of outputs, and instability in the adversarial training process.

Another problem is that Generative Adversarial Networks (GANs) require significant computational resources due to their complex architectures and the need for extensive training on large datasets. Training GANs involves iterative optimization processes between the generator and discriminator, demanding substantial processing power and memory. This computational intensity can limit accessibility, slow down development cycles, and increase costs, especially for those without access to high-performance hardware. This problem could lead to several consequences including:

- Limited Accessibility
- Increased Costs
- Extended Development time

Researchers are continually developing new techniques and architectures to address these challenges, such as Wasserstein GANs (WGANs) and Progressive GANs.

Looking ahead, the future of GANs is promising, with ongoing research exploring new applications and improving existing techniques. Advances in hardware, such as more powerful GPUs and specialized AI accelerators, will further enhance the training and deployment of GANs. As GANs continue to evolve, they are likely to play an increasingly

important role in various industries, driving innovation and enabling new technologies.

Architecture of GANs

The GAN works by having two parts: a generator and a discriminator.

- **Generator:** The generator is a neural network that tries to create new data samples. It starts with a random noise vector and uses it to generate an output that resembles the target data distribution.
- **Discriminator:** The discriminator is another neural network that tries to determine whether a data sample is real or fake (i.e., generated by the generator). The discriminator is trained on a set of real data samples. Over time, the generator gets better at creating realistic data samples, and the discriminator gets better at telling real from fake [4].

The training process is iterative. Here's a breakdown of the steps involved:

1. **Noise Input:** The generator starts with a random noise vector as input. This noise vector can be thought of as a starting point for the generation process.
2. **Generator:** The generator uses the noise vector to generate a new data sample. This data sample is initially rough and will be refined over multiple training iterations.
3. **Discriminator:** The discriminator receives both real data samples (from the training set) and the generated data samples from the generator. The discriminator's task is to classify each data sample as either real or fake.

4. **Feedback Loop:** The discriminator's output (correct classification or incorrect classification) is then fed back to both the generator and the discriminator themselves. This feedback helps them improve their performance in the next iteration.
5. **Generator Update:** Based on the discriminator's feedback, the generator is updated to improve its ability to generate more realistic data samples.
6. **Discriminator Update:** The discriminator is also updated based on the feedback, so it becomes better at distinguishing between real and fake data samples.
7. **Fine Tune Training:** This iterative process of training the generator and discriminator continues until both reach a point where the discriminator has difficulty classifying the generated data samples as fake. At this point, the generator is considered to be well-trained and can be used to generate new data samples.

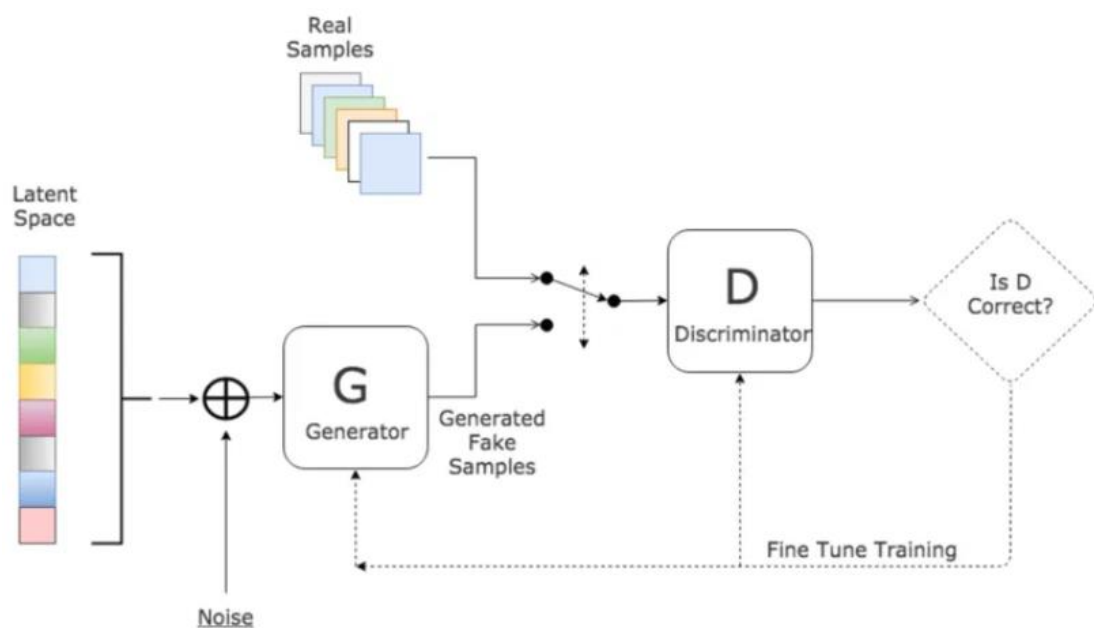


Figure 7. General Architecture of GANs [28]

Mathematical details about GANs

Neural network modeling essentially requires defining two things: an architecture and a loss function. We have already described the architecture of Generative Adversarial Networks. It consists in two networks [5]:

- a generative network $G(\cdot)$ that takes a random input z with density p_z and returns an output $x_g = G(z)$ that should follow (after training) the targeted probability distribution
- a discriminative network $D(\cdot)$ that takes an input x that can be a “true” one (x_t , whose density is denoted p_t) or a “generated” one (x_g , whose density p_g is the density induced by the density p_z going through G) and that returns the probability $D(x)$ of x to be a “true” data.

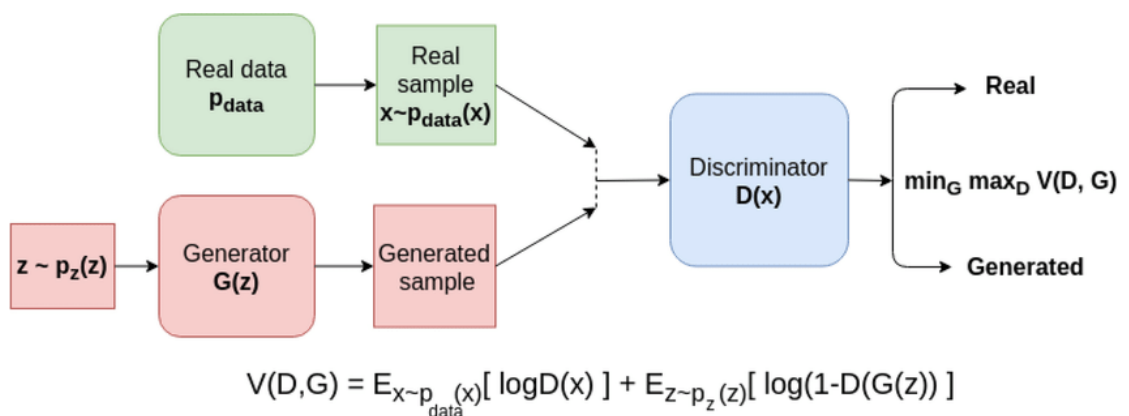


Figure 8. A standard GAN architecture and loss function [5].

Types of GANs

The core GAN architecture has been extended to address various data generation tasks. Here are some prominent variations:

- **Conditional GAN (CGAN):**

While Generative Adversarial Networks (GANs) are impressive in their ability to create new data, they often lack control over the specific type of data generated. Enter Conditional Generative Adversarial Networks (CGANs), a powerful extension that injects control into the generation process. The Core Idea of CGANs build upon the vanilla GAN architecture by incorporating an additional input, typically a label or text description, alongside the random noise vector fed to the generator [6].

- **Deep Convolutional GAN (DCGAN):**

Powering Realistic Image Generation, While Generative Adversarial Networks (GANs) have revolutionized data generation, they often struggle with capturing the intricacies of complex data like images. Deep Convolutional GANs (DCGANs) address this challenge by leveraging the power of convolutional neural networks (CNNs) in both the generator and discriminator. **The benefits of DCGANs Improved Image Quality** By leveraging CNNs, DCGANs generate images with significantly higher quality, detail, and realism compared to vanilla GANs. **Efficient Training:** The use of convolutional layers and stride/fractional-stride convolutions allow DCGANs to process image data efficiently, leading to faster training times [7].

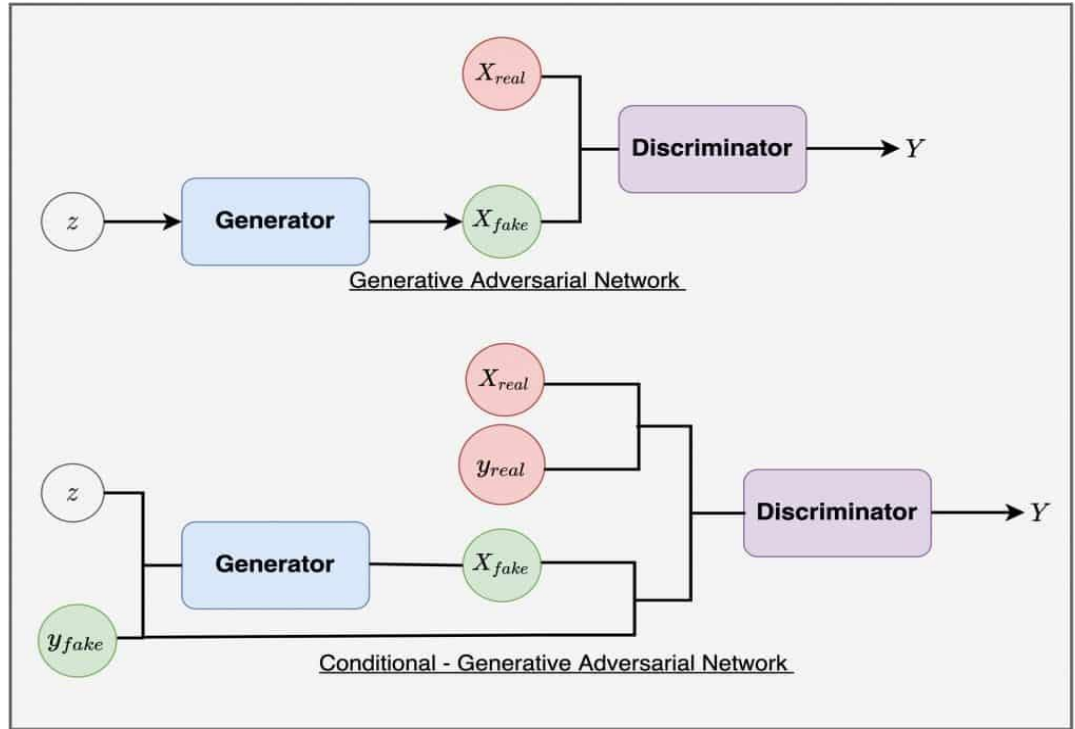


Figure 9. GANs Vs CGANs [14].

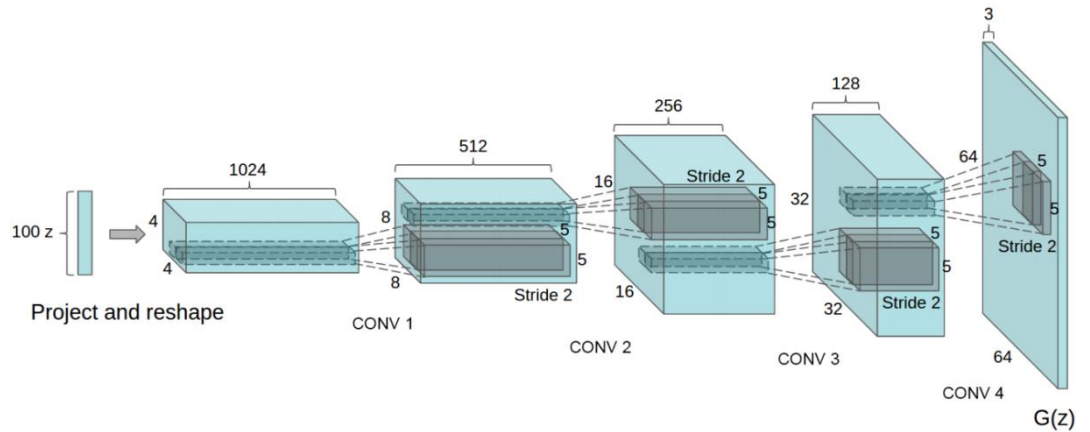


Figure 1: DCGAN generator used for LSUN scene modeling. A 100 dimensional uniform distribution Z is projected to a small spatial extent convolutional representation with many feature maps. A series of four fractionally-strided convolutions (in some recent papers, these are wrongly called deconvolutions) then convert this high level representation into a 64×64 pixel image. Notably, no fully connected or pooling layers are used.

Figure 10. Deep Convolutional GAN Architecture [15].

- **Pix2Pix GAN:**

Pix2Pix GAN, introduced by Isola et al [8], is a type of Conditional Generative Adversarial Network (CGAN) specifically designed for paired image-to-image translation tasks. It has gained immense popularity due to its ability to generate high-quality translations between diverse image pairs, such as converting grayscale images to color, sketches to photographs, or maps to aerial views. Pix2Pix GAN builds upon the CGAN architecture, incorporating an additional input, typically a label or image mask, alongside the random noise vector fed to the generator. This allows the generator to generate images that are conditioned on the provided input, ensuring that the translations are meaningful and relevant to the input image.

- **CycleGAN:**

Unpaired Image-to-Image Translation with a Twist While Pix2Pix GAN excels at image-to-image translation with perfectly paired data obtaining such data can be a challenge. CycleGAN, introduced by Zhu et al. [9], tackles this limitation by enabling image-to-image translation even when paired training data is unavailable. CycleGAN learn a mapping between these domains so you can translate images from one domain to the other[23].

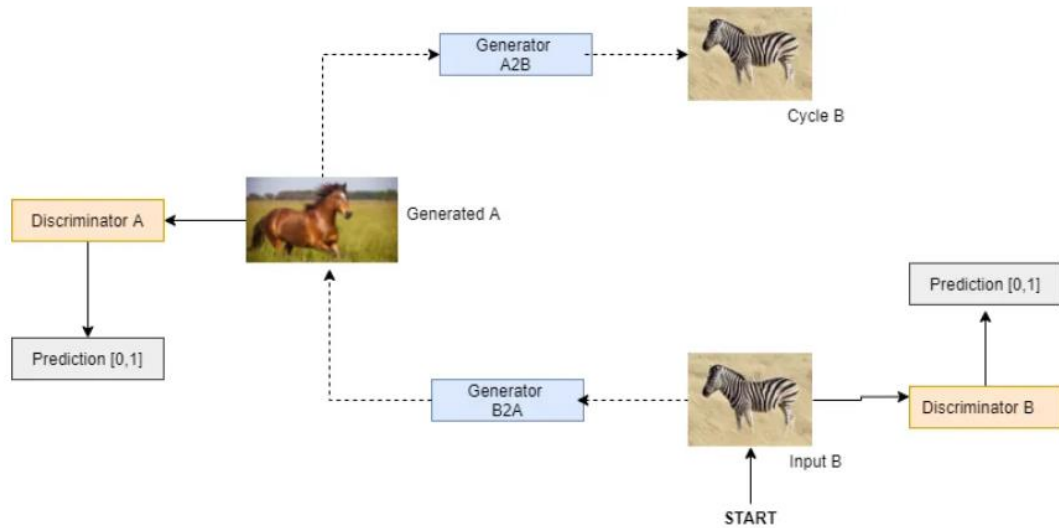


Figure 11 cycleGAN operates under the assumption that you have unpaired datasets from two domains (e.g., horse images and zebra images). The goal is to

1.4.2 CNN (Convolutional Neural Network) Model

Convolutional Neural Networks (CNNs) are a class of deep learning algorithms that are particularly effective for tasks involving spatial data, such as image and video recognition. CNNs are designed to automatically and adaptively learn spatial hierarchies of features through backpropagation by using multiple building blocks, such as convolution layers, pooling layers, and fully connected layers [10].

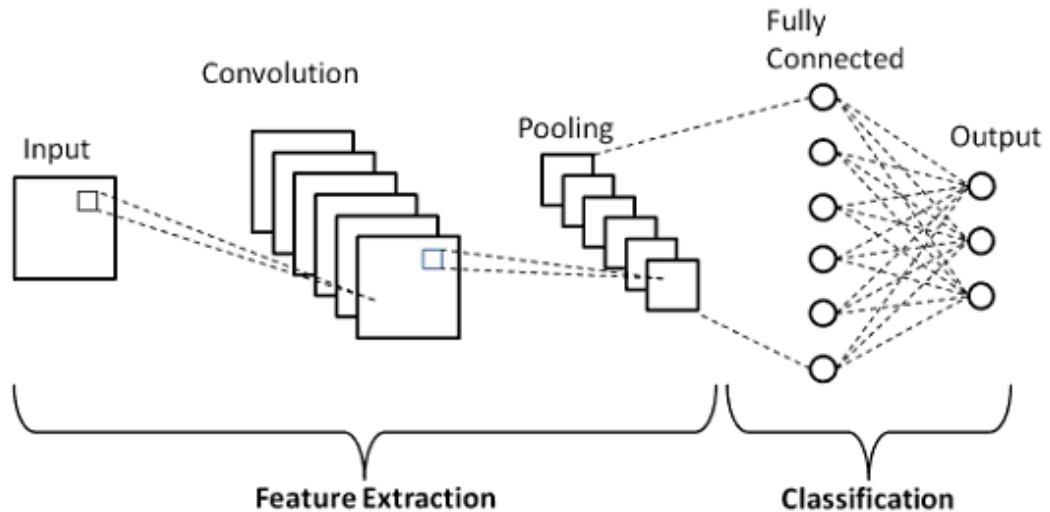


Figure 12. General Architecture of CNN

Applications of CNN

The main advantage of using CNN is that CNN can itself learn features from the input removing the task of manual feature engineering. It is also more efficient in terms of memory and complexity than traditional fully connected neural network. Some of the applications of CNN are: Image Recognition, Image Classification, Speech Recognition, and Text Classification .

The use of CNN models has grown even higher recently, including in cloud-based applications. As the technology is inclining towards cloud computing we assume that CNN will become more popular and its usage will grow more and more [10].

1.5 Project Scope

1.5.1 Description of the Task

In 2022, the Transportation Security Administration (TSA) in the USA began adding facial recognition technology to the security process at airports across the country. The move is part of a larger effort to use biometrics in airport security, which will include facial scanning. Airport facial recognition aims to help the TSA more accurately identify known threats and prevent them from entering secure areas while making it easier for passengers to get through security efficiently. In addition to making travel safer, this technology will reduce wait times at airports by reducing human errors.



Figure 13. New Facial Recognition system by American Airlines and Thales [23]

Aging is an essential contributing factor that can affect error rates in such systems, It has been recently shown that Generative Adversarial Networks (GANs) can produce synthetic images of exceptional visual fidelity. Where GANs can address again dilemma in face recognition systems.

1.5.2 Problem Definition

A clear, concise description of the issue or challenge that the project aims to address. It sets the stage for the entire project by providing context and outlining the specific problem that your project intends to solve.

- **Security is very Sensitive:** Airport security faces challenges in accurately identifying and verifying individuals, particularly due to facial variations caused by aging. This emphasizes the difficulty security has in keeping up with how people's appearances change.

- **Other biometric technologies are not the best to use.**

Limitation of other biometric technology:

Fingerprints: Can be smudged or obscured by dirt impacting accuracy,

Iris Scans: This may be less comfortable for some users due to the proximity of the scanner to the eye.

Voice Recognition: This can be affected by background noise, accents, or illness.

But In facial recognition:

Ubiquitous data: Faces are readily available people don't need to carry special tokens or remember passwords.

Accuracy: Facial recognition technology has become increasingly accurate, especially with advancements in deep learning.

- **There is an age imbalance in the existing datasets:** The gaps between the ages of people in the same datasets vary from person to another.
- **Machine learning typically didn't solve aging yet:** Traditional models are primarily focused on classification or regression tasks. They wouldn't be able to generate new, realistic images of aged faces, which can be helpful for tasks like visualizing the aging process or creating training data for other models.

1.5.3 Motivation

The motivation of a project explains why the problem is worth solving. It provides the rationale behind the project and emphasizes its importance and potential impact. The main objective of this project is to study the Generative Adversarial Networks (GANs) model to address the face Aging problem.

So, our steps in this work:

- Study and analyze various GAN models for face aging recognition.
- Test various GAN models for face aging recognition.
- Trying to design a hybrid classification algorithm.

Chapter 2

Planning

2.1 Introduction

Project planning plays a crucial role in the successful execution of any research project. It helps define clear goals, establish a roadmap for achieving those goals, and manage resources effectively. In the context of our graduation project on face aging recognition using a Generative Adversarial Network (GAN), a well-defined plan will ensure we systematically approach each stage of development, from data collection to model evaluation. This planning chapter outlines the key steps involved in our project, including task breakdown, scheduling, and resource allocation. By following this roadmap, we aim to stay organized, address potential challenges efficiently, and ultimately achieve the desired outcomes of our research.

2.2 The Planning Process

The planning process as adopted by most planners is often based on the theory of rational comprehensive planning theory. The typical planning process is a dynamic process starting from identifying issues, to stating goals and objectives and then collecting and interpreting data.

After data collection and analysis plans are prepared, adopted, and implemented. During the implementation process, new issues might arise, therefore calling for an upward iterative process to improve planning efficiency and effectiveness. [11]

In our work, we followed this cycle as follows:

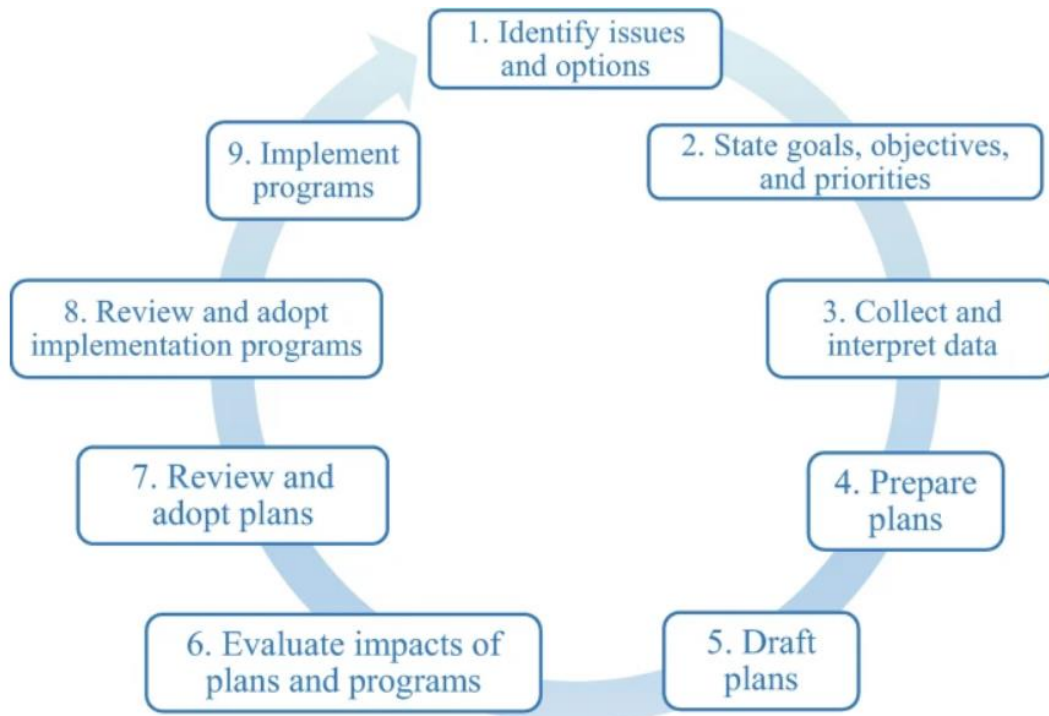


Figure 14 Planning Process Cycle [11]

1. **Identify Issues and Options** We identified the specific problems in the face aging recognition field and explored various options for solving them. **Issues** Difficulty in generating realistic aged face images, and challenges in recognizing faces across different age groups. **Options** Use of GANs like CycleGAN or StarGAN for image generation, and implementation of advanced face recognition algorithms.
2. **State goals, objectives, and priorities:** after looking at the problem definition, we thought about objectives and what we should do to solve these problems. **Goals** Develop a system to generate aged face images using GANs and accurately recognize faces across different age groups. **Objective** Ensure high-quality image generation, improve face recognition accuracy, and validate the system with real-world datasets. **Priorities** Focus first on image generation, thenonintegrating recognition algorithms, followed by rigorous testing and evaluation.

- 3. Collect and interpret data:** In this step, we collect data (images) to train and evaluate the model. And finding the dataset that we need to use and that is appropriate for the scope of our project.

Analyze the dataset to understand the distribution of age groups, quality of images, and potential preprocessing requirement.
- 4. Prepare Plans:** Review the models used in previous related work, and compare the results to achieve the suitable plan that we will follow.

Plan Development Create a comprehensive project plan that includes timelines, resources, and task assignments. **Technical Planning** Outline the technical steps required for data preprocessing, model training, evaluation, and system integration.
- 5. Draft Plan:** the decision of model selection that suits our work, and takes into consideration Tasks required for each stage (e.g., data cleaning, training the GAN model, evaluating performance). Resource allocation (e.g., software tools, computing power).

Developing documentation for each phase of the project, including algorithms, system architecture, and testing protocols.
- 6. Evaluate the impacts of plans and programs:** Milestones to track progress (e.g., completion of data pre-processing, achieving a specific accuracy on a validation set). Timelines for each task and milestone.

Impact Assessment: Evaluate the potential effectiveness of the chosen GANs and recognition algorithms. **risk analysis:** Identify risks such as overfitting, and data biases.
- 7. Review and adopt Plans:** We made adjustments to the plan as we needed to address potential issues or ensure alignment with our goals.

Once we were satisfied, formally we adopted the plan to guide our project execution.

- 8. Review and adopt implementation:** Ensure these programs have clear steps to achieve the goals outlined in the plan. Consider potential challenges that might arise during implementation and plan mitigation strategies.
- 9. Implement Program:** Train GANs to generate aged face images and fine-tune face recognition models. System Integration: Integrate the image generation and face recognition components into a cohesive system. Testing and Validation: Perform extensive testing and validation to ensure the system meets the defined goals and objectives.

2.3 Project Tasks

2.3.1 TASK ASSIGNMENT

Task assignment is crucial for efficiently managing a project, ensuring that each team member knows their responsibilities and can focus on specific areas of the project. Consider using project management tools like Notion, Trello, Asana, or Jira to track progress, assign tasks, and manage deadlines. Regular team meetings and updates can also help ensure everyone is on the same page and any issues are addressed promptly.

Task	Name / Title	Start Date	End Date	Weeks	Assigned	Predecessors
1	Problem Definition	1-10-23	21-10-23	3	1,2,3,4,5,6,7,8	--
2	Defining Objectives	22-10-23	18-11-23	4	1,2,3,4	1
3	Studying prev. apps	22-10-23	18-11-23	4	5,6,7,8	1
4	Full Analysis	19-11-23	30-12-23	6	1,2,3,4,5,6,7,8	2,3
5	Software Design	31-12-23	2-3-24	9	1,2,3,4,5,6,7,8	4
6	System Implementation	3-3-24	30-3-24	4	1,2,3,4	5
7	Selecting best Performance	3-3-24	23-3-24	3	5,6,7,8	5
8	System Testing	31-3-24	21-4-24	3	1,2,3,4,5,6,7,8	6,7
9	System Documentations	22-4-24	1-6-24	6	1,2,3,4,5,6,7,8	8
10	Project Revision	2-6-24	21-6-24	3	1,2,3,4,5,6,7,8	9
Total				38		

Figure 15 Task Assignment from the Allocation

2.3.2 GANTT CHART

A Gantt chart is a valuable tool in project planning and management, providing a visual timeline for tasks and milestones. It displays project activities along a timeline, illustrating their start and end dates, durations, and dependencies. Each task is represented by a bar, with the position and length reflecting the task's schedule.

By clearly outlining the sequence of activities and highlighting critical paths, Gantt charts facilitate efficient resource allocation, better communication among team members, and enhanced overall project coordination. This structured approach aids in identifying bottlenecks, enabling proactive adjustments to keep the project on track and within scope.

Gantt chart of project planning:

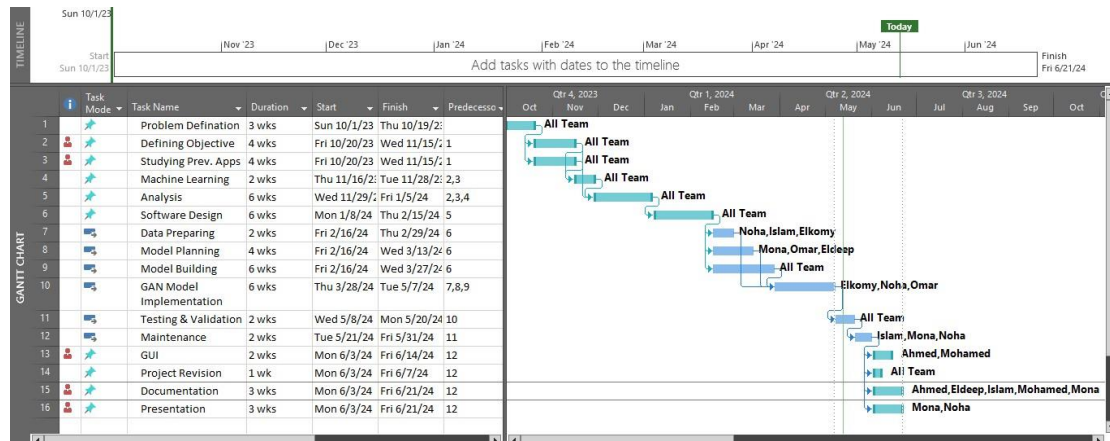
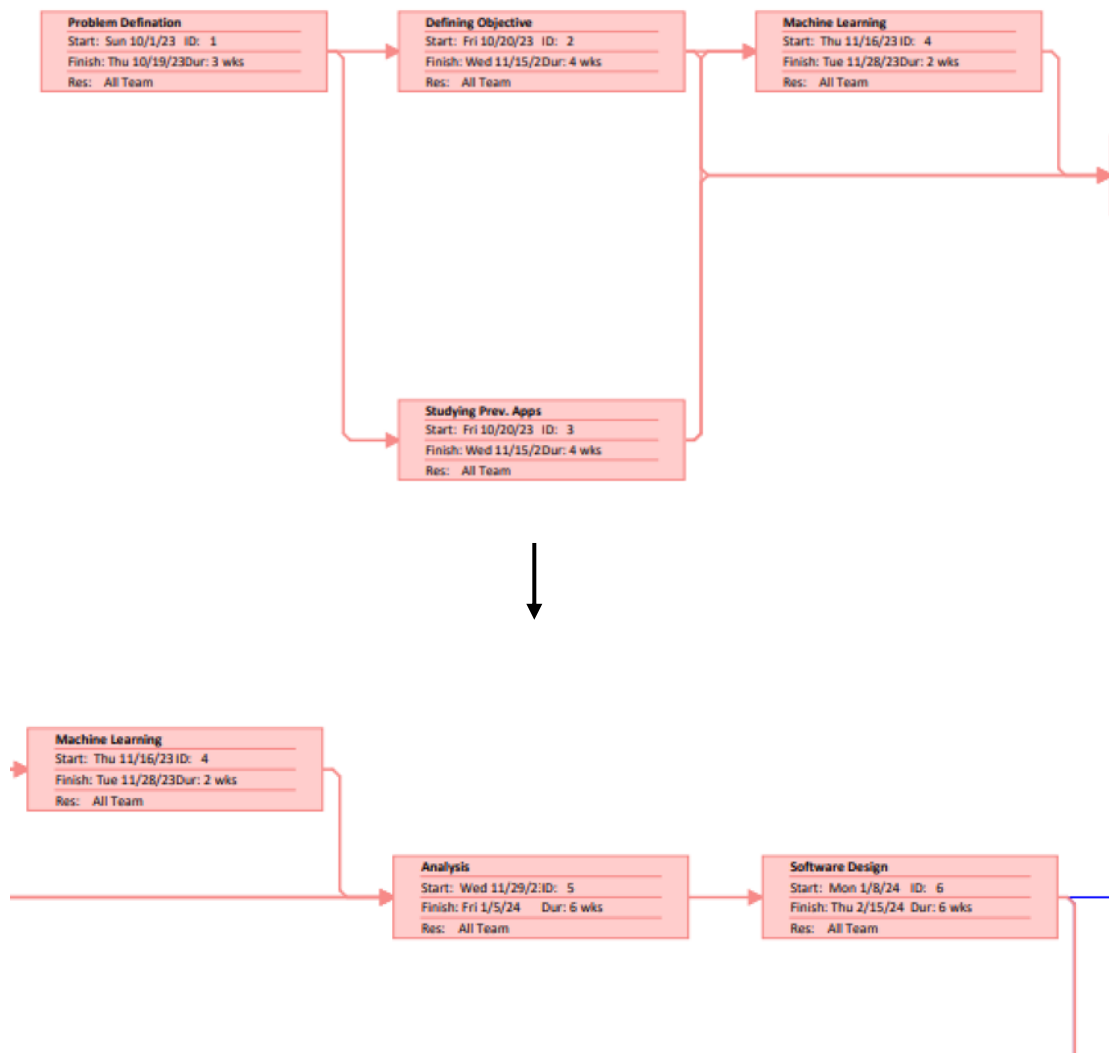


Figure 16 Gantt Chart of Project Planning

2.3.3 NETWORK DIAGRAM

A Network diagram is another essential tool in project management, offering a visual representation of a project's activities and their interdependencies. Unlike a Gantt chart, which focuses on timelines, a Network diagram emphasizes the sequence and relationships between tasks. It typically consists of nodes (representing activities) connected by arrows (indicating dependencies). This layout helps in identifying the critical path, the longest sequence of tasks that determines the project's duration. By highlighting task dependencies, Network diagrams facilitate better planning, resource allocation, and risk management, ensuring that potential bottlenecks are addressed proactively to keep the project on schedule.

Chapter 2: Planning



Chapter 2: Planning

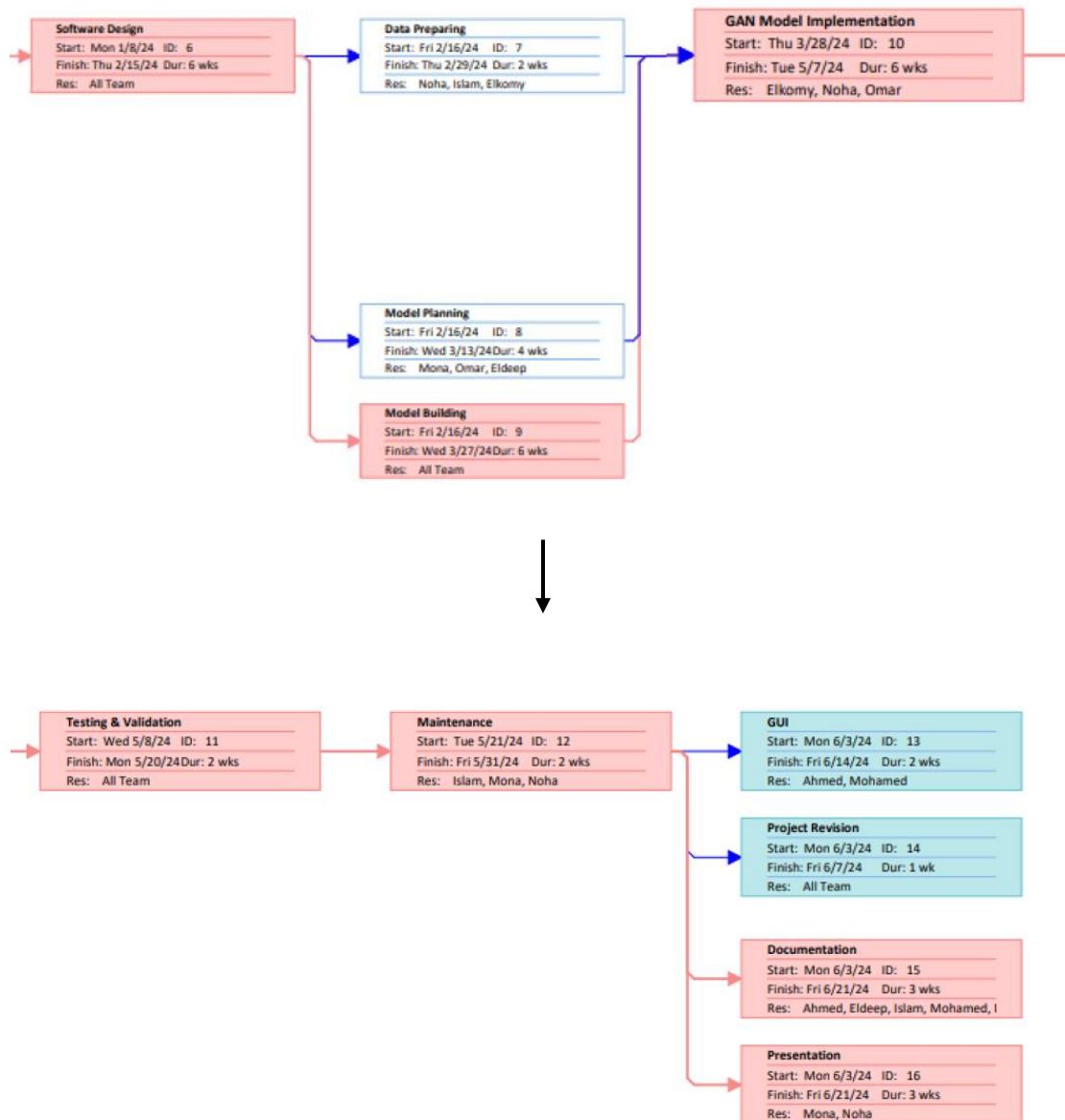


Figure 17 Network Diagram of project planning

Software Tools Used in Project Planning

Drow.io: a versatile and user-friendly online diagramming tool, allows project managers to construct detailed network diagrams that clarify the relationships between activities. By dragging and dropping shapes [Figure 10].



Figure 18 Draw.io

Microsoft Project: a comprehensive project management software, allows users to create detailed Gantt charts that display the start and end dates, durations, and dependencies of project tasks [Figure 11].



Figure 19 MS Project

Chapter 3

Literature Review

3.1 Artificial Intelligence Overview

Artificial intelligence (AI), developed in 1956, is a field of informatics that through complex systems, allows machines to learn from experience, mimicking the capabilities of the human mind [12]. In particular, AI can process and manipulate large amounts of data and extrapolate correlations between them in a way the human brain would not recognize (Figure 12). AI's application fields are innumerable and have revolutionized and improved the management of many activities [12].

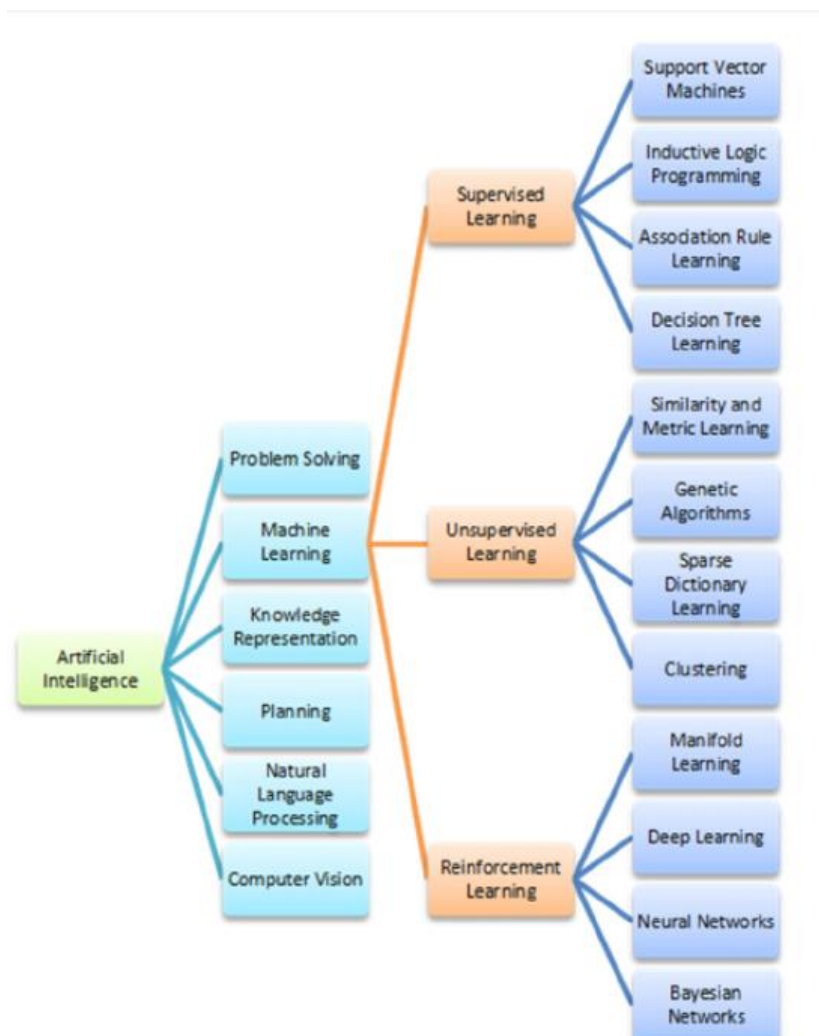


Figure 20 AI main fields are depicted in Figure12. Machine learning is the heart of

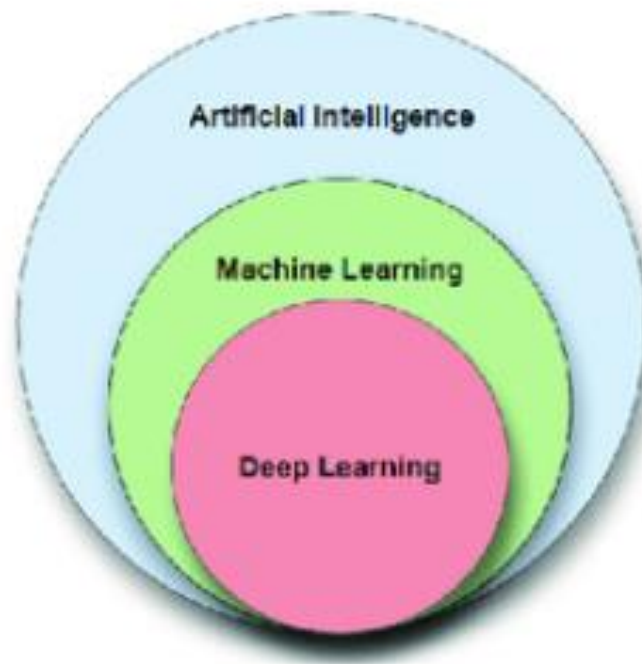


Figure 21. AI main fields

Artificial Intelligence (AI): is the broad field of creating machines capable of performing tasks that typically require human intelligence. These tasks include reasoning, learning, problem-solving, perception, language understanding, and interaction.

Machine Learning (ML): ML is a subset of AI that focuses on developing algorithms that allow computers to learn from and make predictions or decisions based on data. It involves training a model on a dataset to recognize patterns and make decisions.

Deep Learning (DL): DL is a subset of ML that uses neural networks with many layers (deep neural networks) to analyze various types of data. These layers progressively extract higher-level features from the raw input, making DL particularly powerful for complex tasks like image and speech recognition.

AI main fields are depicted in (figure 12) Machine learning is the heart of AI, and its more promising research area is represented by

so-called deep learning, an evolution of the artificial neural network basic approach.

Machine learning (ML) is focused on the creation of AI programs that learn or improve performance based on the data they use. It enables algorithm creation that can learn and make predictions [12].

Deep learning (DL) is an algorithm type that uses a multi-layer neural network architecture to learn representations of data.

3.2 Face Recognition Concepts

Face recognition is the process of identifying people through facial images. It has become vital for security and surveillance applications and is required everywhere including institutions, organizations, offices, and social places [6]. There are several challenges faced in face recognition which include face pose, age, gender, illumination, and other variable conditions. Another challenge is that the database size for these applications is usually small. So, training and recognition become difficult [6].

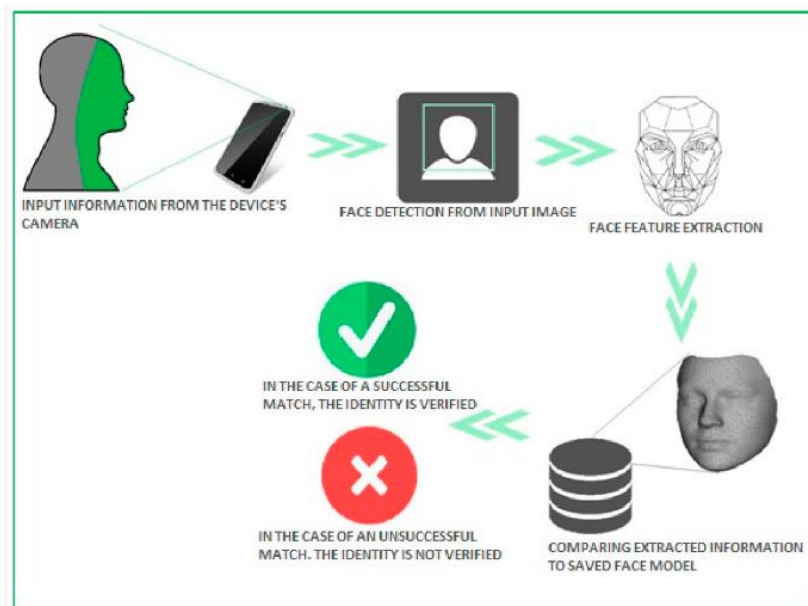


Figure 22 . Face-recognition system schema

3.2.1 Face Recognition Block Diagram

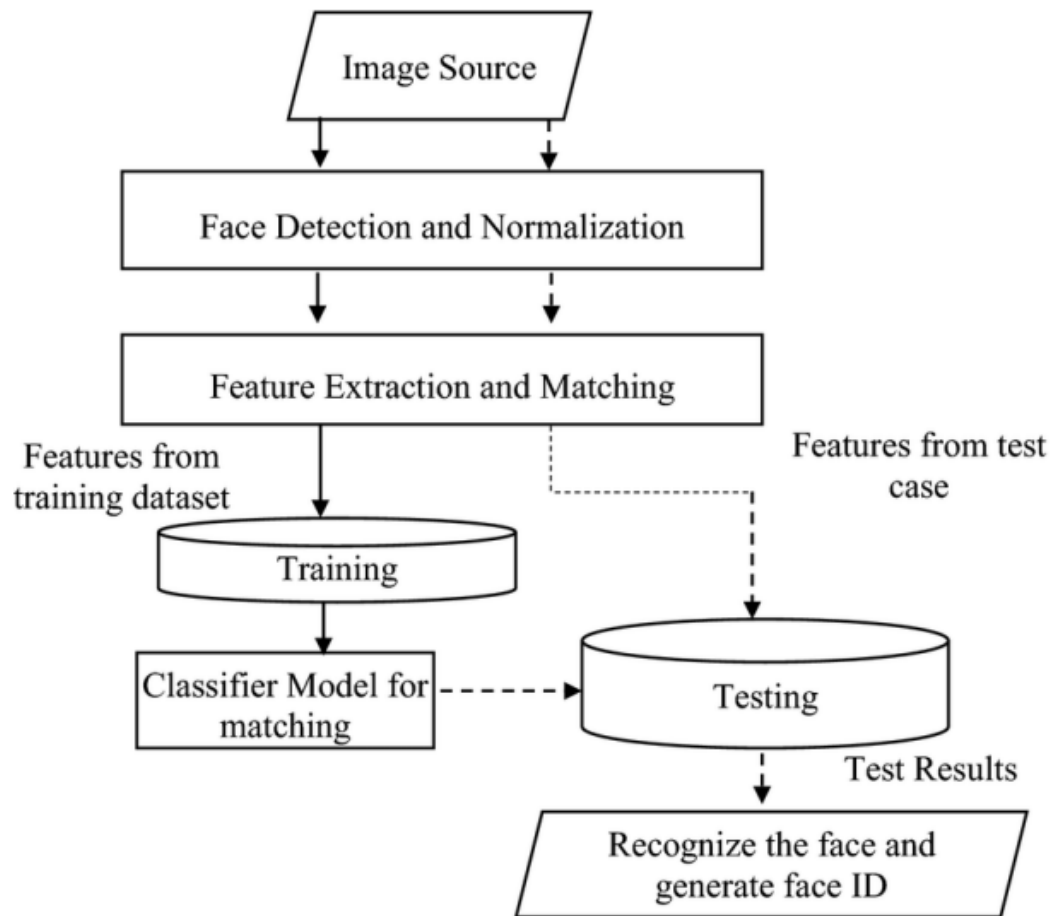


Figure 23 Block diagram of the human face recognition system [6]

1. **Image Source** (Acquiring facial images)

The image source is the initial step where the system acquires the facial images to be processed. This could be from various inputs such as a camera, image database, or a real-time video feed.

2. **Face Detection and Normalization** (Identifying and aligning faces)

This step involves identifying and isolating the face within the image and normalizing it for further processing. Face Detection Techniques like Haar Cascades, Histogram of Oriented Gradients

(HOG), or deep learning-based methods (e.g., MTCNN, YOLO) detect faces within an image.

Normalization involves aligning the detected face to a standard size and orientation, correcting for variations in lighting, and removing background noise. Normalization ensures that the face is in a consistent state for feature extraction.

3. **Feature Extraction and Matching** (Creating and comparing feature vectors)

distinctive features of the face are extracted to create a unique representation (feature vector) of the face, some algorithms used for feature extraction are:

- Local Binary Patterns(LBP)
- Scale-Invariant Feature Transform (SIFT)
- CNN

4. **Training** (Building a classifier model using training data)

This phase involves using the features extracted from a labeled training dataset to train a classifier model. Training Dataset is a collection of images with known identities and their corresponding feature vectors. Classifier Models are Machine learning models such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), or deep learning models like CNNs are trained to learn the mapping between feature vectors and identities. Training Process The model learns to classify faces based on the extracted features by adjusting its parameters to minimize classification errors.

5. **Classifier Model for Matching** (Using the trained model for face recognition)

This is the trained model that is used to match extracted features from new, unseen images to known identities. The classifier takes the feature vector of a test face as input and predicts the identity by comparing it against the learned feature vectors from the training phase. The accuracy and performance of this model are crucial for reliable face recognition.

6. **Testing** (Evaluating the model with test data)

The testing phase involves evaluating the classifier model using test cases to ensure its effectiveness. **Test Case Features** Features are extracted from new images that were not part of the training dataset. **Testing Process** The classifier model predicts the identity of the face in the test image and the results are compared against the ground truth to measure accuracy .

7. **Recognize the Face and Generate Face ID** (Identifying and labeling the face)

The final step where the system identifies the face and generates the corresponding face ID. **Recognition** is The classifier's output is used to recognize the face and determine the identity. **Face ID** is A unique identifier or label is assigned to the recognized face, which can be used for various applications such as security verification, attendance systems, or personalized user experiences.

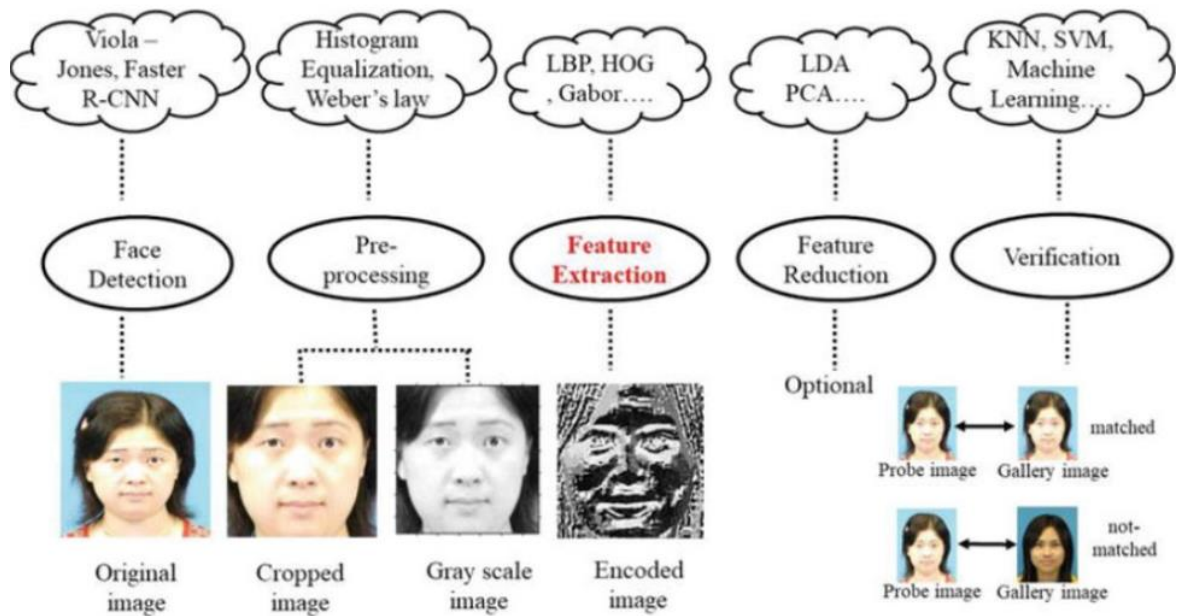


Figure 24 A complete structure of face recognition system. [26]

3.2.2 Pre-processing Faces

Pre-processing faces refers to a series of steps and techniques applied to facial images before they are used for tasks like face recognition, facial expression analysis, or any other face-related computer vision tasks. The main objective is to prepare the raw facial images in such a way that they become suitable for effective and accurate analysis by machine learning algorithms [13].

The series of steps:

1. **Face Detection:** A cascade-based face detector is used to locate the face in an image. This step gives a rough estimate of the face's scale and position, allowing us to resize the face to a standard size.
2. **Identifying Facial Features:** To achieve accurate face recognition, we need to identify distinct features on the face, such as the eyes, nose, mouth, etc. We aim to identify 20 key points on

the face, including the corners of the eyes, the tip of the nose, the nostrils, and the corners of the mouth.

3. **Warping the Face:** Once the features are identified, we use triangulation to map these features to a standard face shape. This process involves warping the image so that the relative positions of these key points match a standard face template. The color of each pixel is preserved during this warping process.
4. **Feature Extraction:** After warping, we need to extract detailed features from the face for recognition. A regular grid of 25 squares is placed around each facial feature point, and the average gradient (in eight directions) and mean intensity in each square are calculated. This is done for each color channel, resulting in a feature vector for each point [13].
5. **Dimensionality Reduction:** To manage the high-dimensional feature vectors, they are projected into a lower-dimensional space (from 675 to 100 dimensions) using training data. This helps retain the most important variations that are common across different poses.
6. **Improving Feature Detection:** Detecting these facial features accurately can be challenging due to variations in individual appearances, expressions, and occlusions (e.g., facial hair). To improve accuracy, we use a Gaussian mixture model trained on a dataset of faces. This model helps predict the probability of a feature's position based on the local image data. Additionally, we use a Gaussian model to represent the relationships between feature positions, ensuring that certain features (e.g., eyes) are not found in illogical positions (e.g., below the mouth). This model is simplified to a tree structure for computational efficiency [13].

7. **Dynamic Programming for Feature Search:** To efficiently find the best positions for all features, dynamic programming is used on the tree structure of features. This method is robust and can handle changes in facial expressions, blinking, and facial hair [13].
8. **Final Feature Extraction:** Once the facial features are accurately detected and the face is warped, the final feature extraction is performed. This involves measuring gradients and intensity levels in specified regions around each feature point, which are then used for face recognition.

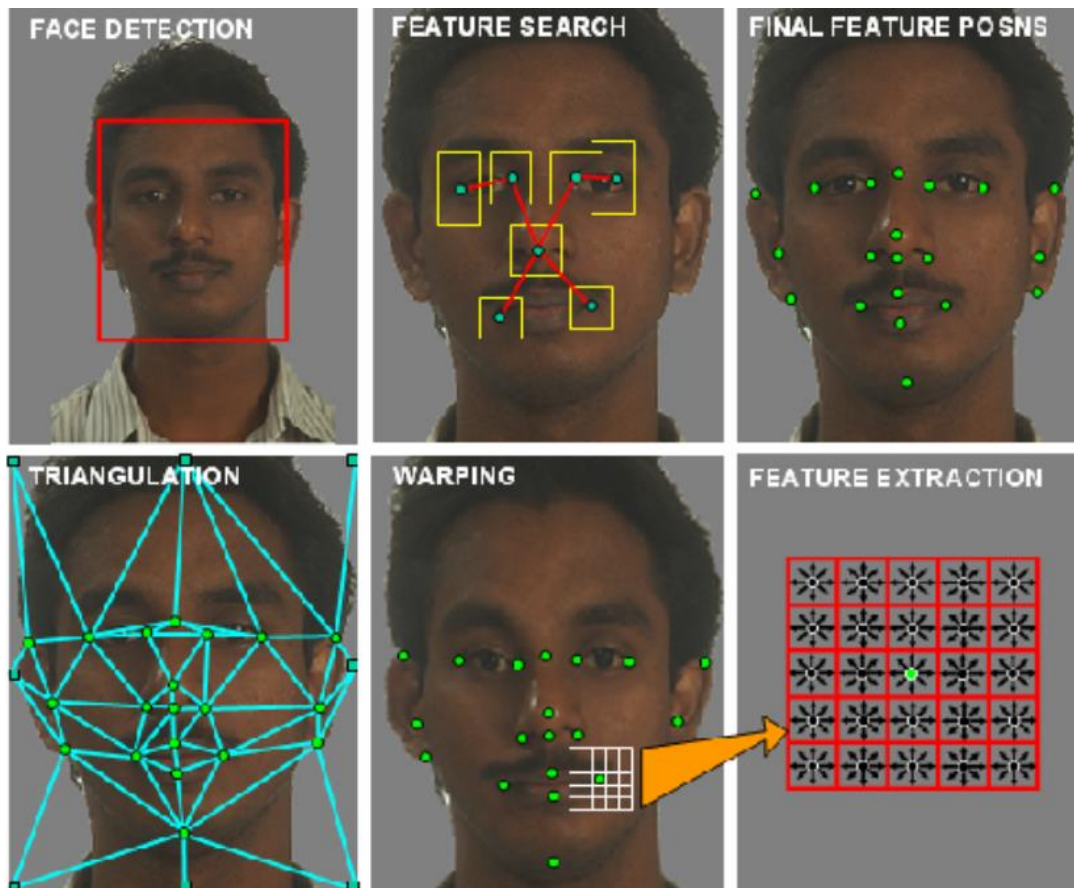


Figure 25 Preprocessing for faces [13]

3.3 Age-Invariant Face Recognition

Age progression is the process of aesthetically rendering a given face image to present the effects of aging. It is often used in the entertainment industry and forensics, e.g., forecasting the facial appearances of young children when they grow up or generating contemporary photos of missing individuals.

The intrinsic complexity of physical aging, the interferences caused by other factors (e.g., PIE variations), and the shortage of labeled aging data collectively make face age progression a rather difficult problem. The last few years have witnessed significant efforts in tackling this issue, where aging accuracy and identity permanence are commonly regarded as the two underlying premises of its success. The early attempts were mainly based on the skin's anatomical structure and they mechanically simulated the profile growth and facial muscle changes w.r.t. [14].

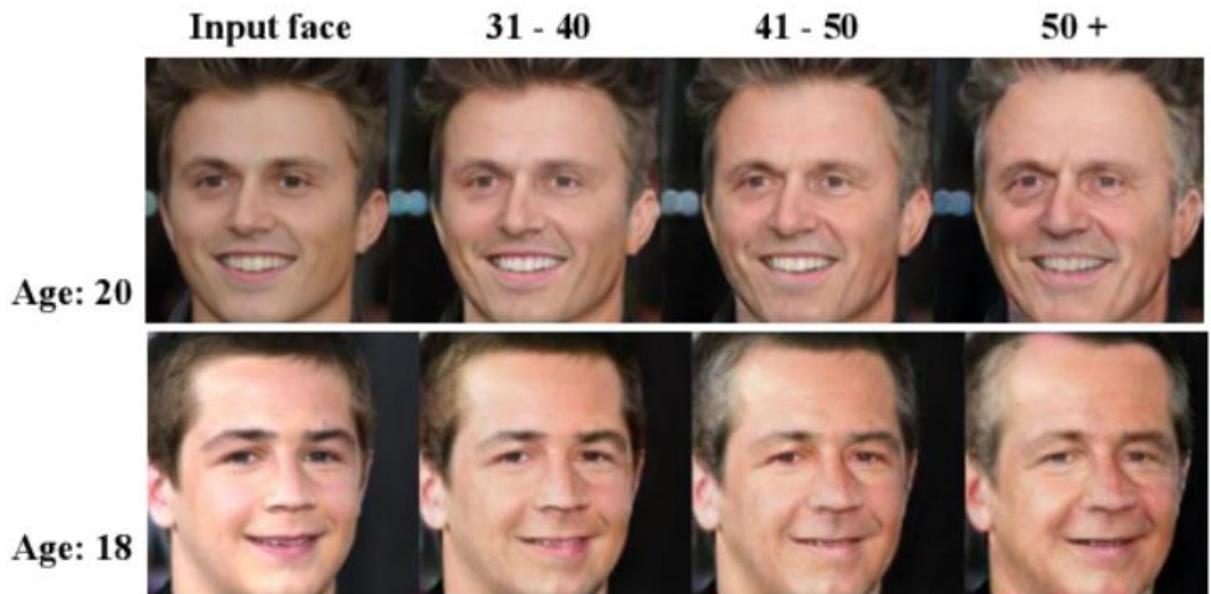


Figure 26 Demonstration of our aging simulation results (images in the first column are input faces of two subjects) [14]

These methods provided the first insight into face aging synthesis. However, they generally worked in a complex manner, making it difficult to generalize. Data-driven approaches followed, where face age progression was primarily carried out by applying the prototype of aging details to test faces or by modeling the dependency between longitudinal facial changes and corresponding Although obvious signs of aging were synthesized well, their aging functions usually could not formulate the complex aging mechanism accurately enough, limiting the diversity of aging patterns [14].

3.3.1 Age-Invariant Face Recognition

Age-Invariant Face Recognition (AIFR) refers to the technology and methods used to identify or verify individuals' faces regardless of the changes that occur in their appearance due to aging. Different from general face recognition, age-invariant face recognition (AIFR) aims at matching faces with a big age gap. Previous discriminative methods usually focus on decomposing facial feature into age-related and age-invariant components, which suffer from the loss of facial identity information [15].

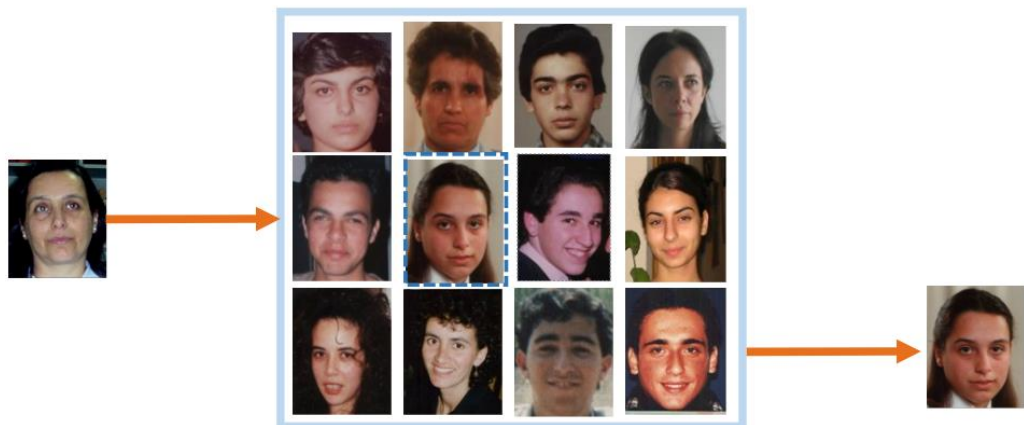


Figure 27 Age-Invariant Example "FG-NET Data Set"

3.4 Related Work

3.4.1 Related Paper Comparison

Title and Year of the Paper	Authors	Dataset	Techniques Used	Results		
Face aging with Conditional generative adversarial network, 30 May 2017	Grigory Antipov Moez Baccouch e Jean-Luc Dugelay	1. Age-cGAN 2-IMDB-Wiki 3-IMDB Wiki cleaned	1-Age estimation CNN	Reconstruction type	Accuracy	
				Initial Reconstruction	53.2%	
				"Pixelwise" Optimization	59.8%	
				"Identity-Preserving" Optimization	<u>82.9%</u>	
A Study on the Effect of Ageing in Facial Authentication and the Utility of Data Augmentation to Reduce Performance Bias Across Age Groups, 6 September 2023	Wang Yao , Muhammad Ali Farooq Joseph Lemley And Peter Corcoran	1 -AgeDB 2-Morph II	1- FR model.	AgeDB is a better option for studying the impact of age, as it includes a considerable quantity of image samples for each individual and is more uniformly spread across various age groups.		
Aging effects in automated face recognition, August 2016	Miguel Cedeno Agamez	1.VGGFace 2.OpenFace 3.FaceNet	1-PCA 2-LDA 3-EBGM		Age-range	Accuracy
				VGGFace	18-25	92.2%
					65-75	72.5%
				OpenFace	18-25	88.3%
					65-75	69.1%
				FaceNet	18-25	<u>95.1%</u>
65-75	78.6%					

Title and Year of the Paper	Authors	Main points	Models
A survey of face recognition Dec 2022,	Xinyi Wang, Jianteng Peng, Sufang Zhang, Bihui Chen, Yi Wang, Yandong Guo	1. History of Face Recognition . 2. pipelines in deep learning framework. 2.1 Preprocessing 2.2 Testing and Training a FR deep model 2.3 Face Recognition by Comparing face embedded . 3. Algorithms that Used in Recognition. 4. Datasets and comparisons.	1- FR deep model
Face Age synthesis July 2023	Ayşe Kale, Oğuz Altun	1-data sets used in face age synthesis 2- methods used in face age synthesis 2.2 classical methods 2.3 deep learning models	1. CNNs 2. RNNs 3. GANs
USING BIOMETRICS FOR FACIAL RECOGNITION AT AIRPORTS December 2022	Juraj Horkay Samer Al-Rabeei Volodymyr Tymofiiiv Technical University of Kosice - Technicka univerzita v Kosiciach	1. BIOMETRICS FACIAL RECOGNITION 2. THE USE OF BIOMETRICS TODAY:- 3. FACIAL RECOGNITION APPLICATION TESTING:- 4. CONCLUSION	Convolutional Neural Networks CNNs

Title and Year of the Paper	Authors	Main points	Models
The use of biometric technology at airport 13 november 2021	Nimra khan , marina Efthymiou	1-Biometric definition , 2-why biometric technology used in airports , 3-biometric identifiers , 4-advantage and disadvantages of biometric, 5-the uses of biometric	1-Minutiae-based models 2-template based models , 3-eigenface, fisherface, 4-local binary pattern (LBP) model 5-hidden markov model
Feasibility Study of a New Security Verification Process Based on Face Recognition Technology at Airport 2020	Taotao Zhu and Lijun Wang	1. Intelligent security verification process 2. Empirical Test and analysis of the intelligent security verification process	1-processed by the IDEA (International Data Encryption Algorithm)
Applications of Gans july 12 2019	Jason Brownlee PhD	1-List of most common Gans applications	1-Generative Adversarial Networks GANs

3.4.2 Main Points and conclusions of Each Related Paper

Face aging with Conditional generative adversarial network, 30 May 2017

In this work, they have proposed a new effective method for synthetic aging of human faces based on ages conditional generative adversarial network (Age-cGan). The method is composed of two steps

- input face reconstructions requiring the solution of an optimization problem in order to find an optimal latent approximation.
- face aging itself performed by a simple change of condition at the input of the generator the cornerstone of our method is the novel (identity-preserving) latent vector optimization approach allowing to preservation of the original person's identity in the reconstruction. This approach is universal meaning that can be used to preserve identity not only for face aging but also for other face alterations (e.g. adding a beard, sunglasses etc.)

A Study on the Effect of Ageing in Facial Authentication and the Utility of Data Augmentation to Reduce Performance Bias Across Age Groups, 6 September 2023

This paper investigated comprehensive studies of the aging effect and facial aging datasets. In addition to this, the impact of age group on the performance of face recognition algorithms is explored, along with an initial exploration of the use of synthetic age data to complement the real datasets. These experiments demonstrate that older people are more accurately identified, whereas younger individuals are more difficult to identify.

To minimize the effect of data imbalance from the original dataset on these results, they employed synthetic age data to expand the number of samples for each age group. This further verifies that the face recognizer exhibits age-specific biases and it is more likely to distinguish between two older people and less likely to confuse them as being the same person. Among the datasets used in this study, this research illustrates that the AgeDB is a better option for investigating the impact of age, as it encompasses a considerable quantity of image samples for each individual and is more uniformly spread across various age groups, with a large interval between the minimum and maximum ages.

Aging effects in automated face recognition, August 2016

Aging impacts the accuracy of the system, The error rate increases as the age gap between the images widen, The deterioration in the performance is due to substantial changes in facial appearance that occur with aging, Face recognition systems need to be able to accommodate age variations to ensure reliable identification across different stages of an individual's life

The study only considers a limited range of age differences(18-25 &65-75) , The study only uses three face recognition algorithms and it does not consider the impact of other factors on face recognition.

A survey of face recognition Dec 2022.

The application of FR we introduce here is face generation, especially for the ones with ID preserving. We divide face generation methods into three types: GAN, 3D, and residual learning based face generation. As we introduced before, algorithms in subsection have merged face generation

and recognition together. A lot of applications, such as face editing (age/expression changing, glasses/bread removing), face swapping use GAN to generate synthetic face images. 3D based methods usually generate faces with different angle, namely face frontalization and rotation. Residual learning based methods usually focus on generating faces without much content changing, such as face denoising, deblurring, super resolution. However, many face denoising and super resolution problems adopt GAN to model the problem.

Face Age Synthesis July 2023

AGAN , developed in this context, accepts conditional information by embedding face attribute information in both the generator and the discriminator. They utilized the attention mechanism to improve visual accuracy, preserve image detail, and limit changes in age-related areas. Unlike other studies on attention modules, They also used Wavelet packet transform to capture textural features and improve aging details by transforming the input images into the frequency domain. MORPH and CACD datasets are used for training, and FGNET and CelebA are used for testing Ref : Y. Liu, Q. Li. Z. Sun. T. Tan, AGAN: An Attribute-Aware Attentive Generative Adversarial Network for Face Aging, IEEE Trans. Inf. Forensics Secur. 16 (2021)

Tools and methods for performance evaluation

1. Face++ The online application Face++ is frequently used to estimate the ages of synthetically generated faces. The estimated age can be used in the training and testing of models

2. FaceNet FaceNet is a 140-million-parameter network that generates a 128-dimensional feature vector from face images and is used to demonstrate the effectiveness of the proposed facial age synthesis methods. It learns to map face images directly into Euclidean space and performs face recognition, validation, and clustering tasks.
3. OpenFace OpenFace is an open-source facial recognition solution that decides whether two face images belong to the same person or not. It is used in the literature to determine identity preservation.

The use of biometric technology at airport 13 november 2021

Image preprocessing techniques:

- Normalization: adjust image for consistent lighting conditions and contrast to ensure uniformity in biometric conditions.
- Noise Reduction: employing algorithms or filters to reduce noise, such as salt and pepper or distortion to improve accuracy of feature extraction. divide into specific region of interest, isolating spatial features for more effective analyses.
- Rotation and scale normalization: ensure that the image are consistently oriented and scaled to facilitate accurate comparison and scaling.
- Feature extraction: Fingerprint Recognition Minutiae Extraction: This technique involves identifying and extracting minutiae points, such as ridge endings and bifurcations, from a fingerprint.

Algorithms then use the relative positions and orientations of these points to create a unique fingerprint

- template Facial
- Recognition:

Geometric Features: Key facial features, such as the distance between the eyes, nose shape, and the length of the jawline, can be extracted to create a facial template. Texture Analysis: This involves analyzing the unique texture patterns of facial regions, such as the eyes, nose, and mouth, for identification.

Iris Recognition:

- Iris Feature Extraction: The unique patterns within the iris, such as furrows, freckles, and coronas, are extracted and used to create a distinctive template for identification.

Retina Recognition:

- Vascular Pattern Extraction: Retina recognition focuses on the blood vessel patterns in the retina. These patterns are extracted and used as features for identification.

Using Biometrics for Facial Recognition at Airports

December 2022

- **Models** Centralized Verification Model: This model is responsible for verifying the identity of passengers at the self-service verification point. The model uses a deep learning algorithm to compare the passenger's facial image to their stored image in the system.
- Automatic Recheck Model: This model is responsible for re-verifying the identity of passengers as they pass through the security screening gate

Preprocessing Techniques

- Image normalization: This technique is used to adjust the brightness, contrast, and color of facial images to ensure consistency and comparability.
- Face cropping: This technique is used to extract the facial region from the image, excluding the background and other irrelevant parts.
- Face alignment: This technique is used to align the facial features in the image to a standard template, making it easier for the facial recognition model to identify and compare faces.
- Face landmark detection: This technique is used to detect and extract the key facial landmarks, such as the eyes, nose, and mouth. These landmarks are used to represent the facial features and aid in facial recognition.

Feature Extraction

- possibility is to use deep learning algorithms to extract the unique features from facial images

Dimensionality Reduction

- possibility is to use principal component analysis (PCA), which is a popular dimensionality reduction technique that is often used for image data. PCA works by identifying the most important features in the data and then projecting the data onto a lower-dimensional subspace that is spanned by these features. This can significantly reduce the number of parameters in the model and improve its computational efficiency.
- Another possibility is to use t-distributed stochastic neighbor embedding (t-SNE), which is a more recent dimensionality reduction technique that is well-suited for preserving the relationships between the data points. t-SNE works by trying to map the data points from a high-dimensional space to a low-dimensional space while preserving the distances between the data points. This can be useful for facial

recognition applications, as it can help to ensure that the model is able to distinguish between different individuals.

Feasibility Study of a New Security Verification Process Based on Face Recognition Technology at Airport Models 2020

The paper proposes a two-stage security verification process based on face recognition technology. The first stage, pre-screening, uses a deep learning model to identify individuals who are likely to pose a security threat. The second stage, verification, uses a traditional face recognition model to confirm the identity of the individual. The paper does not specify the specific models that are used in each stage. However, it is likely that deep learning models are used in both stages, as they have been shown to be effective for face recognition tasks. Based on my understanding of the paper, the following models are likely to be used:

- **Pre-screening:** A convolutional neural network (CNN) is likely to be used for pre-screening. CNNs are well-suited for image classification tasks, such as identifying individuals who pose a security threat.
- **Verification:** A support vector machine (SVM) is likely to be used for verification. SVMs are well-suited for binary classification tasks, such as confirming the identity of an individual.

Preprocessing Techniques The paper does not specify any preprocessing techniques that are used. However, it is likely that standard techniques such as image cropping, resizing, and normalization are used. These techniques are used to prepare the images for input to the face recognition models. They help to

ensure that the images are of a consistent size and format, and that they are not corrupted or distorted.

Feature Extraction The paper does not specify the specific features that are extracted. However, it is likely that features that are invariant to changes in pose, illumination, and expression are used. These features are extracted from the images to represent the facial features of the individual. They are used by the face recognition models to compare the input image to a database of known images.

Dimensionality Reduction The paper does not specify whether dimensionality reduction is used. However, it is likely that dimensionality reduction is used to reduce the number of features, which can improve the performance of the face recognition models. Dimensionality reduction techniques such as principal component analysis (PCA) can be used to reduce the number of features while preserving the most important information. This can help to improve the accuracy of the face recognition models without sacrificing performance.

A Longitudinal Study of Automatic Face Recognition 2015

They presented a longitudinal study on face recognition, utilizing a database (PCSO LS) of 147,784 face images of 18,007 subjects. Each subject has at least 5 images which were acquired over at least a 5 year time interval. Multi- level statistical models were used to estimate population- mean trends in genuine scores, particularly with respect to increasing elapsed time between two face images. Based on the results of our longitudinal analysis, our findings are summarized as follows:

1. Significantly decreasing trends in genuine scores over time were

observed for the two state-of-the-art COTS face matchers, with COTS-A showing overall better performance than COTS B.

2. Despite decreasing genuine scores, population-mean trends of COTS-A genuine scores from all demographic groups remained above the threshold at 0.01% FAR for time intervals up to 16 years (the maximum in the PCSO LS database):

3. The probability of false rejection by COTS-A was stable (at less than 2%) across time intervals up to approximately 10 years for 0.01% FAR.

4. Some of the variation in subject-specific estimated parameters was explained by gender, race, and face quality covariates. Face quality measures did not explain the variation in genuine scores as well as time interval alone, but improved the model fit when included with time interval.

Face aging simulation with a new wrinkle oriented active appearance model 25 June 2018

we presented a new framework to analyze facial aging taking into account shape, appearance and wrinkles. We showed that the system can generate realistic faces for aging and rejuvenating, and such age-progressed faces better influence age perception than with Active Appearance Model or Conditional Adversarial Autoencoder. On average, we demonstrated an improvement factor of 2.0 over prior works [63, 27]. Nevertheless, the model can be improved in several ways. Firstly, the realism of the faces produced by the model has not been rated in this study. Moreover, we know that facial aging is influenced by environmental factors like sun exposure, alcohol consumption or eating practices. A potential improvement could be to compute multiple trajectories in function of those factors. In addition, dark spots must be

included in the model to increase the accuracy of facial aging. We are confident that dark spots can be integrated in the same way as wrinkles. This is the objective of future research.

3.4 Artificial Intelligence Models in Facial Aging

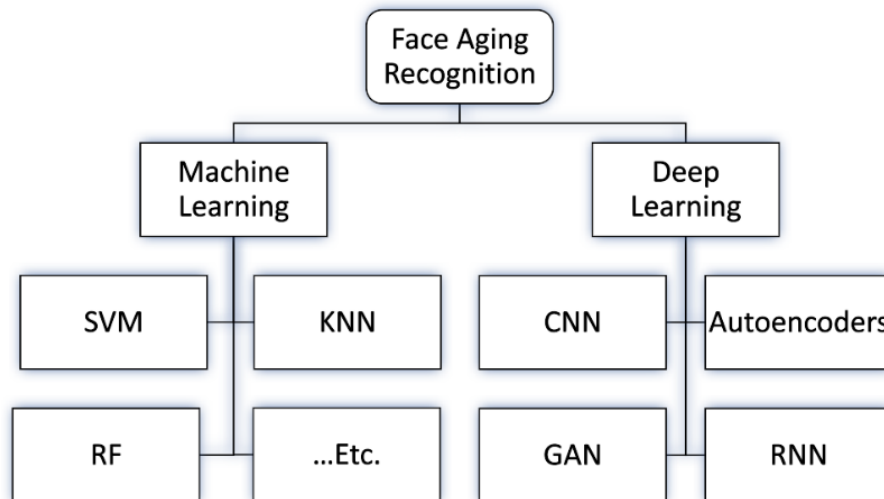


Figure 28. ML and DL Models

Models And Techniques We Used in Our Experiments:

Decision Tree Algorithm

A decision tree is a flowchart-like structure used to make decisions or predictions. It consists of nodes representing decisions or tests on attributes, branches representing the outcome of these decisions, and leaf nodes representing outcomes or predictions. Each internal node corresponds to a test on an attribute, each branch corresponds to the result of the test, and each leaf node corresponds to a class label or a continuous value [16].

Concepts and Components:

Root Node Represents the entire dataset and the initial decision to be made. Internal Nodes Represent decisions or tests on attributes. Each internal node has one or more branches.

Branches Represent the outcome of a decision or test, leading to another node. Leaf Nodes Represent the final decision or prediction. No further splits occur at these nodes.

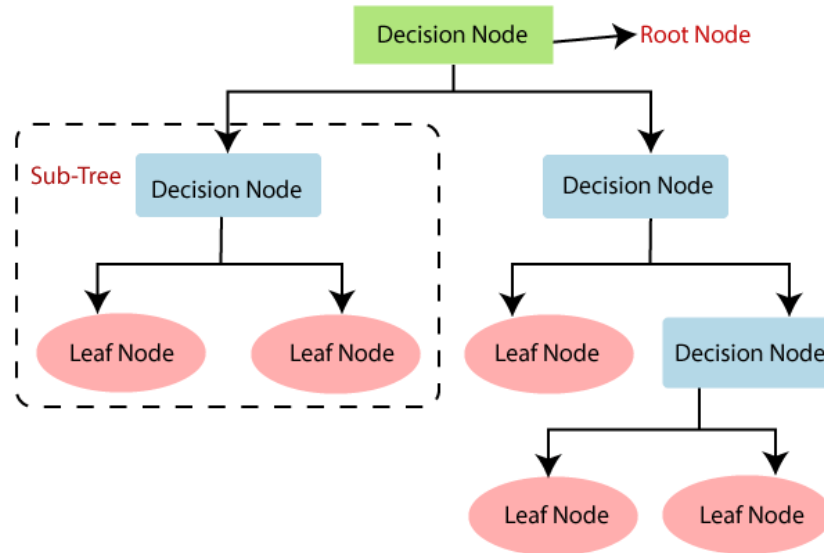


Figure 29. Decision Tree Algorithm in Machine Learning

The Best Attribute is selected using a metric like Gini impurity, entropy, or information gain, the best attribute to split the data is selected. Splitting the Dataset The dataset is split into subsets based on the selected attribute. Repeating the Process The process is repeated recursively for each subset, creating a new internal node or leaf node until a stopping criterion is met (e.g., all instances in a node belong to the same class or a predefined depth is reached) [16].

Advantages of Decision Trees

- **Simplicity and Interpretability:** Decision trees are easy to understand and interpret. The visual representation closely mirrors human decision-making processes.

- **Versatility:** Can be used for both classification and regression tasks.
- **No Need for Feature Scaling:** Decision trees do not require normalization or scaling of the data.
- **Handles Non-linear Relationships:** Capable of capturing non-linear relationships between features and target variables.

Applications of Decision Trees

- **Business Decision Making:** Used in strategic planning and resource allocation.
- **Healthcare:** Assists in diagnosing diseases and suggesting treatment plans.
- **Finance:** Helps with credit scoring and risk assessment.
- **Marketing:** Used to segment customers and predict customer behavior.

Support Vector Machine (SVM)

SVM is a supervised machine learning algorithm used for classification and regression tasks. It finds the hyperplane that best separates different classes in the feature space.

Types of Support Vector Machines Based on the nature of the decision boundary, Support Vector Machines (SVM) can be divided into two main parts:

Linear SVM: Linear SVMs use a linear decision boundary to separate the data points of different classes. When the data can be precisely linearly separated, linear SVMs are very suitable. This means that a single straight line (in 2D) or a hyperplane (in higher dimensions) can entirely divide the data points into their respective classes. A hyperplane that maximizes the margin between the classes is the decision boundary.

Non-Linear SVM: Non-Linear SVM can be used to classify data when it cannot be separated into two classes by a straight line (in the case of 2D). By using kernel functions, nonlinear SVMs can handle nonlinearly separable data. The original input data is transformed by these kernel functions into a higher-dimensional feature space, where the data points can be linearly separated. A linear SVM is used to locate a nonlinear decision boundary in this modified space.

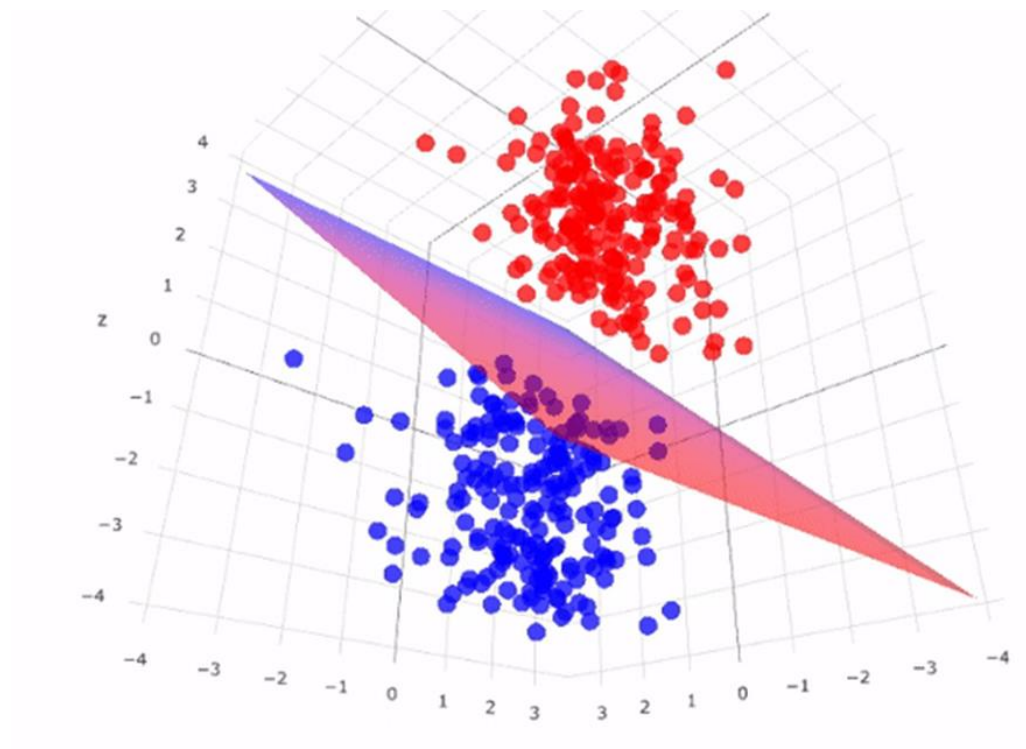


Figure 30. Support Vector Machine (SVM)

Advantages of SVM

- Effective in high-dimensional cases.
- Its memory is efficient as it uses a subset of training points in the decision function called support vectors.
- Different kernel functions can be specified for the decision functions and it's possible to specify custom kernels.

Applications:

- Text and image classification-
- Handwriting recognition-
- Bioinformatics (e.g., cancer classification)

PCA (Principal Component Analysis) Principal Component Analysis (PCA) is a dimensionality reduction technique used to transform a large set of variables into a smaller one while retaining most of the variability in the data. It does this by finding new variables, known as principal components, which are linear combinations of the original variables and are uncorrelated [17].

PCA components and concepts

- **Principal Components** are new variables created from linear combinations of the original variables. Each principal component captures the maximum variance in the data along a new axis orthogonal to the previous components.
- **Variance** is the Measure of the spread of data points. PCA seeks to maximize the variance in the data along the principal components.

- **Eigenvectors and Eigenvalues:** PCA is mathematically grounded in the eigen decomposition of the covariance matrix. Eigenvectors determine the directions of the principal components, while eigenvalues determine their magnitude (importance) [17].

PCA has many types Like:

- **Linear PCA:** Which is the standard form of PCA that assumes linear relationships between variables.
- **Kernel PCA:** It is a nonlinear form of PCA that uses kernel functions to map the data into a higher-dimensional space where linear separation is possible.
- **Sparse PCA:** It Introduces sparsity in the principal components, making them more interpretable.

PCA advantages

- **Dimensionality Reduction:** Reduces the number of variables while retaining most of the information.
- **Noise Reduction:** By discarding less significant components, PCA can help in reducing noise.
- **Computational Efficiency:** Simplifies complex datasets, making machine learning algorithms faster and more efficient.

PCA can be used for many applications including:

- Image Compression
- Face Recognition
- Finance
- Bioinformatics
- Speech Recognition

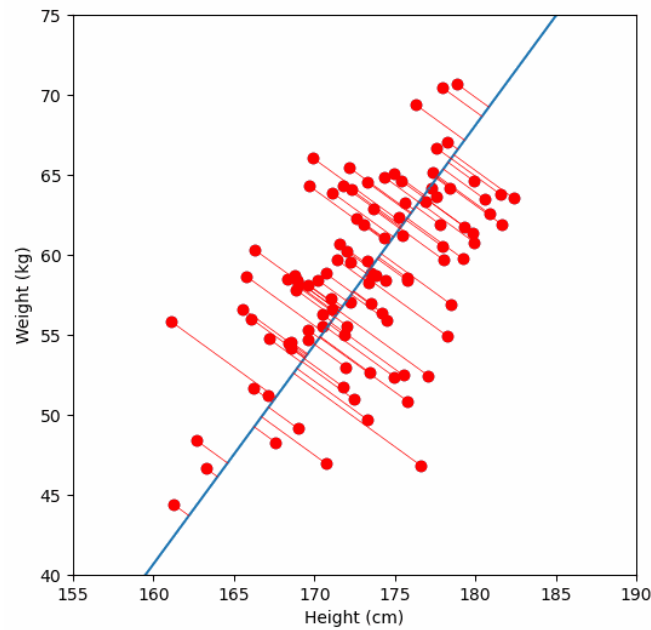


Figure 31.PCA (Principal Component

Wavelet Transformation

Wavelets are mathematical functions used to decompose data into different frequency components and analyze each component with a resolution matched to its scale. Wavelet analysis is a powerful tool for signal processing and is used for tasks such as data compression, noise reduction, and feature extraction.

It works as follows, Transformation: Apply the wavelet transform to the signal, which involves convolving the signal with scaled and shifted versions of the mother wavelet [18].

Decomposition Levels: Break down the signal into multiple levels of detail, where each level captures different frequency components. Inverse Transformation: Combine the decomposed components to reconstruct the original signal.

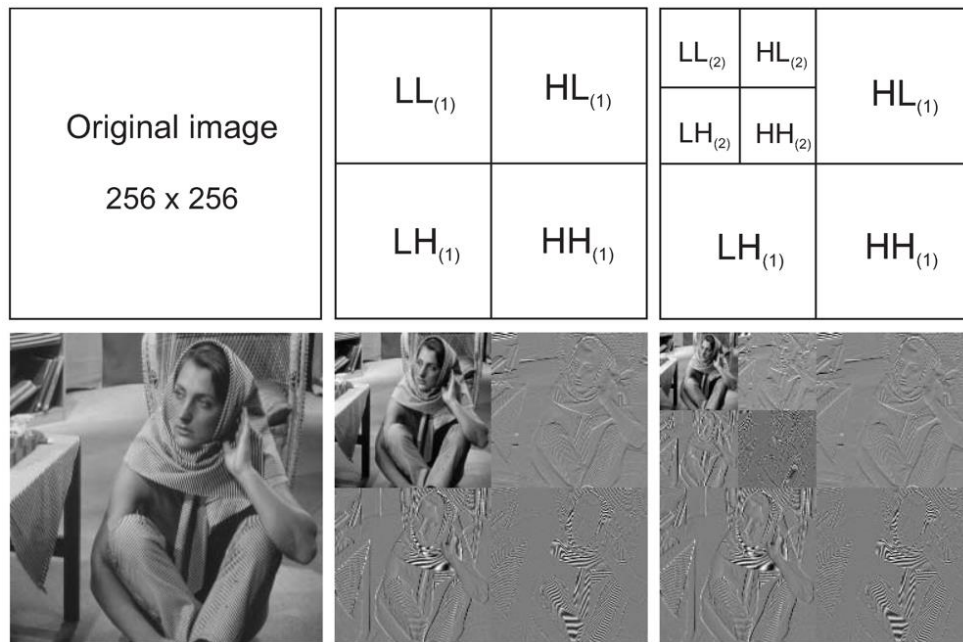


Figure 32.DWT processing generating two and three levels of resolution[11].

Concepts and Components

- **Wavelet Function:** A function used to transform data into different frequency components. Common wavelets include the Haar wavelet, Daubechies wavelets, and Morlet wavelets.
- **Scaling Function:** A function that represents the low-frequency components of the signal.
- **Mother Wavelet:** The original wavelet function from which other wavelets in the family are derived through scaling and translation.
- **Decomposition:** The process of breaking down a signal into its constituent wavelets.
- **Reconstruction:** The process of recombining the wavelet components to recover the original signal.

Wavelet Types

1. **Continuous Wavelet Transform (CWT):** Analyzes the signal at every possible scale and translation, providing a highly detailed time-frequency representation.

2. Discrete Wavelet Transform (DWT): Analyzes the signal at specific scales and translations, offering a more efficient representation with fewer data points.
3. Stationary Wavelet Transform (SWT): A variant of DWT that is invariant to translations of the input signal, useful for denoising and feature extraction.

Advantages

1. Multi-Resolution Analysis: Ability to analyze signals at different resolutions, capturing both time and frequency information.
2. Sparse Representation: Efficient representation of signals, making it suitable for compression and denoising.
3. Localized Analysis: Capability to focus on specific parts of the signal, providing detailed insights.

Applications

1. Signal Processing: Noise reduction, compression, and feature extraction in audio and speech signals.
2. Image Processing: Image compression (e.g., JPEG 2000) and edge detection.
3. Biomedical Engineering: Analysis of physiological signals like ECG and EEG.
4. Finance: Analysis of financial time series for trend detection and anomaly detection.

Chapter 4

Implementation

The aging process varies widely among individuals, posing challenges for face recognition systems reliant on matching current faces to older reference images stored in databases or IDs. Such mismatches can jeopardize security, especially in critical environments like airport security, where faces must be matched not only to scanned passports but also to databases of blacklisted travelers. To address this, our system implements Generative Adversarial Networks (GANs) for face aging and recognition. Integrating GAN-generated aged images and dataset augmentation enhances accuracy and robustness across different age groups, ensuring reliable face verification and identification.

In this chapter, we focus on the practical aspects of implementing our face aging and recognition system using GANs. The implementation phase transforms theoretical concepts into a functional system capable of performing face aging and enhancing recognition accuracy. We generate synthetic aged faces to augment training datasets, improving system robustness. This chapter provides a comprehensive guide on how we implemented the GAN model and integrated it with our face recognition system.

We begin with an overview of the system architecture, describing the interconnection between the GAN and face recognition components. The GAN model augments our training datasets by generating aged facial images, enhancing the system's robustness across different age groups. Next, we explore the intricacies of the GAN model implementation, including the architecture of the Generator and Discriminator networks, the training process, and challenges encountered during development. We also discuss hyperparameter tuning and optimization strategies to enhance performance.

Following the GAN implementation, we focus on the face recognition system, covering feature extraction methods, algorithms for face verification, recognition, and identification, and the integration of augmented datasets into the training pipeline. We highlight advanced techniques for improving accuracy and efficiency in real-world scenarios.

Finally, we discuss the testing and validation of the integrated system, describing the experimental setup, evaluation metrics, and results from various test cases. This comprehensive overview provides insights into the effectiveness of our approach and potential future improvements. By the end of this chapter, readers will have a thorough understanding of the technical implementation of our face aging and recognition system, along with practical insights gained from the development process.

4.1 GANs for Aging

Generative adversarial networks mainly consist of a generator and a discriminator. In this section we're going to talk about the architectures of both the generator and the discriminator and how they were implemented. The architecture of our networks was adopted from J. Y. Zhu et al. [19] who proposed a new concept for unpaired image to image translation, as paired data is not always available hence the previously proposed architectures didn't give great results with unpaired images.

4.1.1 Datasets

We used the **AginG Faces in the Wild(AFGW)** dataset to train our cycleGAN model. It was collected from three resources: the search engine using different keywords, the productive aging laboratory data base and

Mugshot images that are accessible to public domain ohh yeah. It consists of 18,685 images with age ranging between 10 and 1094 years. We only used a subset of the database of 1000 images for each domain, age categories 30-34 and 60-94 were used for domains A and B respectively.

Young Faces(30-34 years old):



Figure 33 AGFW young faces(30-34 years old)

Old Faces(60-94 years old):



Figure 34 AGFW old faces (60-94 years old)

4.1.2 Generator Network Architecture

First, we have the generator, it plays a pivotal role in synthesizing realistic data from random noise, a process fundamental to the GAN framework introduced by Ian Goodfellow et al. [20] in their seminal paper. This generator network architecture consists of 9 residual blocks: c_{7s1-64} , d_{128} , d_{256} , R_{256} , R_{256} , R_{256} , R_{256} , R_{256} , R_{256} , R_{256} , R_{256} , R_{256} , u_{128} , u_{64} , c_{7s1-3} .

It begins with a convolutional layer using a large 7×7 kernel with 64 filters and a stride of 1, capturing detailed spatial features from the input

image. It is followed by two dense layers with 128 and 256 units, respectively, enabling complex feature learning. The core of the network consists of nine residual blocks, each with 256 filters, which facilitate the training of deep networks by allowing gradients to bypass layers, thus justifying the vanishing gradient problem.

For reconstructing a high-resolution output, the network includes two up-sampling layers that incrementally increase the spatial resolution to 128 and then 64 units. Finally, the network concludes with a 7x7 convolutional layer with 3 filters, producing an output suitable for a 3-channel image (such as an RGB image), effectively combining learned features into a coherent and detailed output.

Table 1 Generator Network Architecture

Layer	Configuration	Strides	Output Shape
Conv	c7s1-64	1	(256, 256, 64)
Conv	d128	2	(256/2, 256/2, 128)
Conv	d256	2	(256/4, 256/4, 256)
Conv	R256	-	(256/4, 256/4, 256)
Conv	R256	-	(256/4, 256/4, 256)
Resnet	R256	-	(256/4, 256/4, 256)
Resnet	R256	-	(256/4, 256/4, 256)
Resnet	R256	-	(256/4, 256/4, 256)
Resnet	R256	-	(256/4, 256/4, 256)
Resnet	R256	-	(256/4, 256/4, 256)
Resnet	R256	-	(256/4, 256/4, 256)
Deconv	u128	2	(256/2, 256/2, 128)
Deconv	u64	2	(256, 256, 64)
Conv	c7s1-3	1	(256, 256, 3)

4.1.3 Discriminator Network Architecture

Second, Is the discriminator, it serves as a crucial component in Generative Adversarial Networks (GANs) by acting as a gatekeeper that evaluates the authenticity of generated data. We employ a 70×70

PatchGAN [21] architecture for the discriminator networks,. This type of GAN is designed to classify 70×70 image patches as real or fake, which allows it to effectively capture high-frequency details in the images. The architecture is constructed using a series of 4×4 Convolution-InstanceNorm-LeakyReLU layers, denoted as C_k , where k represents the number of filters, and the layers use a stride of 2. The specific sequence of layers in the discriminator is C64-C128-C256-C512. Notably, the first C64 layer does not utilize InstanceNorm, ensuring that the low-level features are preserved more distinctly. The LeakyReLU activations are used throughout the network with a negative slope of 0.2, providing a slight gradient for negative inputs and preventing the "dying ReLU" problem.

After the final layer, a convolution is applied to yield a one-dimensional output, determining the patch's authenticity. This architecture balances effective feature extraction with computational efficiency, making it suitable for detailed image discrimination tasks.

Table 2 Discriminator Network Architecture

Layer	Configuration	Strides	Output Shape
Conv	C64	1	(256, 256, 64)
Conv	C128	2	(256/2, 256/2, 128)
Conv	C256	2	(256/4, 256/4, 256)
Conv	C512	-	(256/4, 256/4, 256)
Conv	1-Dimensional Output	-	(256/4, 256/4, 256)

4.2 Face Verification and Recognition

For the face Verification and Recognition, we used Googles Facenet pre-trained model. We preferred to use this pretrained model as it was trained on over 200 million pictures and achieve a record high accuracy of 99.63% on the LFW dataset with extra edge detection.

We implemented it using the Deepface library

4.3 User Interface with Tkinter

Integrating the face recognition and verification models with the GUI is a multi-step process involving image capture, data extraction, feature comparison, and user interaction. By connecting these components seamlessly, we create an efficient and user-friendly system that ensures secure and accurate face verification for airport self-check-ins. This integration enhances the overall functionality and reliability of the system, providing a robust solution for modern security needs.

Here is the workflow of the tkinter application:

1. Application Setup:

- **Tkinter Framework:** Develop the desktop application using python's Tkinter, providing a user-friendly interface for airport security personnel.

2. Image Capture and Uploa:

- **CV2 for Image Capture:** Use OpenCV (cv2) to capture images from a webcam or other camera devices. This allows real-time image acquisition for verification.
- **Passport Image Upload:** The system provides an interface for users to upload passport images. A file dialog allows users to

select image files of various formats, including PNG, JPG, JPEG, BMP, and GIF. We used the **open_file_dialog()** function which handles the selection, loading, display and temporary storage of the uploaded image

3. Data Extraction and Display:

- **For the passport picture:** the passport information is extracted from the passport image using **Form Extractor AI** which sends a POST request to a specified API endpoint with the image data and receives a JSON response containing extracted data fields such as name, date of birth, nationality, sex, passport number, and date of expiry.
- **Verification and Decision Making:**
 - **Passport validity is checked:** it is determined and displayed whether this passport is expired or is valid.

Chapter 4: Implementation

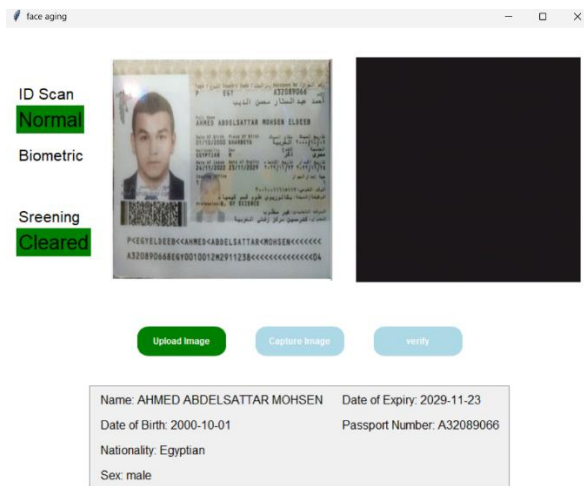


Figure 35 Passport is valid if the person is verified the gates will open

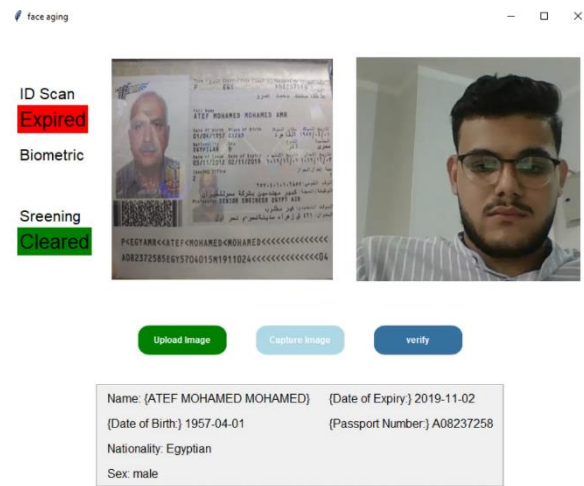


Figure 36 Expired passport would deny access and alert the authorities

- **Data Verification:** Compare the captured image and passport image embeddings using the verification model.
- **Decision Logic:** Based on the verification results and extracted data, decide whether the person can pass. If

verified, display a message and trigger the gate to open.

If not verified, alert the authorities and deny access.

4. **Security and Notifications:**

- **Alert System:** Implement a notification system to alert authorities in case of a mismatch or if the person is denied access. This ensures immediate response to potential security threats.

5. **Workflow**

1. **Start:** The system initializes and the user interface is displayed.
2. **Image Capture:** Capture the image of the person at the self-check-in kiosk using cv2.
3. **Passport Upload:** Upload the passport image and extract the face for recognition.
4. **Data Extraction:** Use an API to extract and display passport data.
5. **Face Verification:** Compare the captured image with the passport image using the FaceNet model.
6. **Decision Making:**

- If the images match, display a success message, open the gate, and check the person in.
- If the images do not match, alert the authorities and deny access.

7. **End:** The process repeats for the next person.

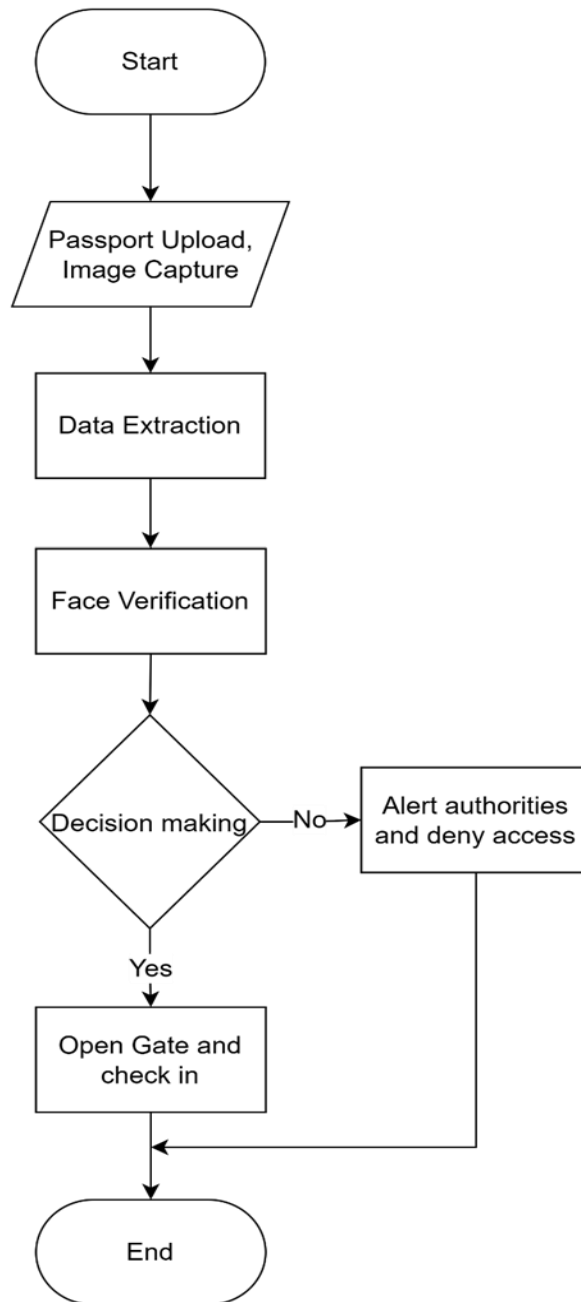


Figure 37 Work flow of face recognition integration in the airport

4.4 Conclusion

In this chapter, we explored the implementation of our face aging and recognition system, focusing on enhancing our datasets using GANs and integrating sophisticated face recognition and verification models. We started by explaining how we utilized CycleGAN to augment our dataset. By generating a diverse set of facial images that span various ages, we ensured our training data was robust, allowing models like FaceNet to recognize faces accurately despite age differences.

Next, we explored the face recognition and verification processes in detail. This included the steps for extracting features from images, generating embeddings, and comparing these embeddings using distance metrics. We also discussed how these models are integrated into a user-friendly GUI developed with Tkinter, enabling users to easily upload and display passport images and verify data.

We then described the workflow for an actual airport self-check-in system. In this scenario, real-time image capture is managed using OpenCV, while scanners are employed for acquiring passport images. This setup ensures that images and data are processed efficiently and accurately, ultimately enhancing both security and the user experience at airport check-in kiosks.

Overall, integrating GANs for dataset augmentation, advanced face recognition models, and an intuitive user interface culminates in a comprehensive system capable of performing reliable age-invariant face recognition. This chapter highlighted how each component works together seamlessly, demonstrating the practicality and effectiveness of our approach

in real-world applications like airport security self-check-ins. By ensuring that every element functions cohesively, we have created a robust solution that addresses the modern needs for security and user convenience.

Chapter 5

Experiments & Testing

Testing, in the context of our project, involves systematically evaluating the performance and reliability of our GAN-enhanced face recognition system. This process ensures that the models function correctly under various conditions and can generalize well to new, unseen data. For our face recognition technology aimed at airport security, testing is crucial to validate the accuracy, robustness, and efficiency of both the GAN-based data augmentation and the face recognition models.

Importance of Testing in Our Project

Testing plays a pivotal role in our project for several reasons:

1. **Validating Model Performance:** Testing allows us to measure how accurately our face recognition system performs. By using a well-defined test set, we can assess key metrics such as accuracy, precision, recall, and F1-score. These metrics indicate how well the model identifies and verifies faces, which is essential for airport security applications.
2. **Ensuring Robustness:** In real-world scenarios, models need to handle diverse and unpredictable inputs. Testing ensures that our models are robust against variations in lighting, facial expressions, angles, and occlusions—common challenges in airport environments.

3. **Identifying and Addressing Issues:** Through thorough testing, we can detect and address any weaknesses or errors in our models. This includes recognizing biases in the data, handling edge cases, and ensuring the system does not fail under specific conditions. Addressing these issues is critical for deploying a reliable and secure face recognition system.
4. **Complying with Regulations:** Airport security systems must comply with various regulatory standards and policies. Testing helps verify that our system meets these standards, ensuring that it can be legally and ethically deployed.
5. **Building User Trust and Acceptance:** For passengers and airport staff to trust and accept the new face recognition system, it must be demonstrably accurate and reliable. Testing provides the evidence needed to build this trust by showcasing the system's effectiveness and reliability.
6. **Ensuring Continuous Improvement:** Testing is not a one-time task but an ongoing process. Regular testing allows us to monitor the performance of our system post-deployment, ensuring it adapts to new data and evolving security requirements.

By thoroughly testing our GAN-enhanced face recognition system, we can confidently deploy a solution that enhances security, improves passenger throughput, and maintains high standards of accuracy and reliability in the dynamic environment of airport security.

Many experiments were held to test the performance of GANs for aging faces as well as the performance of the face verification and recognition system. First set of experiments compare how different parameters affect the output of the GAN model. The second set of experiments compare the result of the SVM, SVM+PCA, and CNN over a face recognition dataset and the dataset after augmentation using the aged faces generated using the GAN model.

5.1 Generative adversarial Networks(GANs)

In this experiment we tested how different parameters affect the output of Generative adversarial networks(GANs) models. We employed CycleGAN for the face aging. The training was done in 2 stages with different parameters and resources. This experiment is focused on the quality of the generated aged and deaged faces.

Dataset

AGFW (Aging Faces in the Wild)

Computer Specification

Cloud Computing Service	
RAM	60Gi Ram
CPU	16core
GPU	T4-4XLarge

Parameters

Cycle-GAN 1st round:

- No of images = 150 for each domain
- Batch size = 1
- Epochs = 50

Cycle-GAN 2nd round:

- No of images = 1000
- Batch size = 5
- Epochs = 20

Results

Result of first round

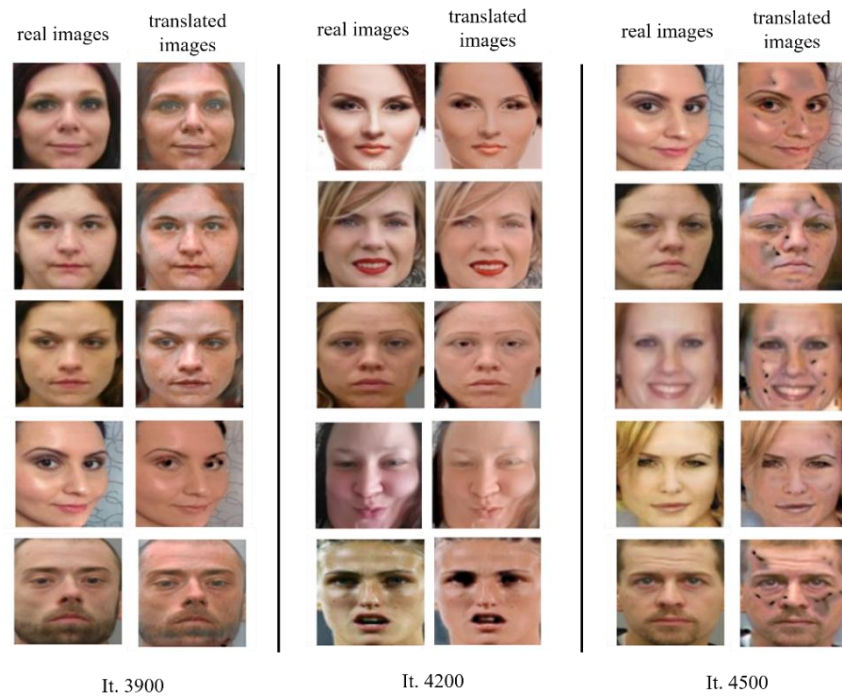


Figure 38 Results during the training of the GAN model at iterations 3900, 4200, 4500

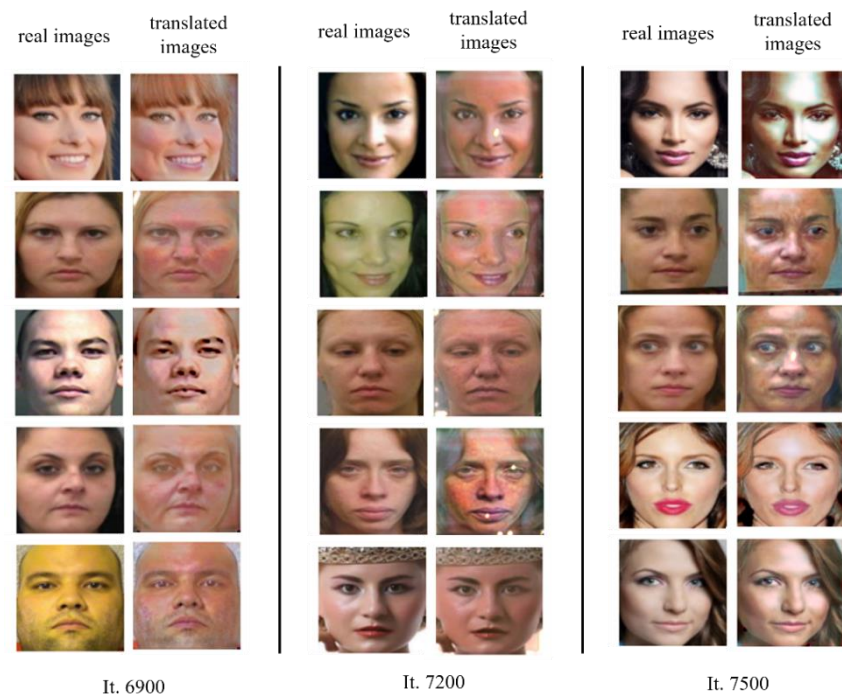


Figure 39 Results during the training of the GAN model at iterations 6900, 7200, 7500

Results of the second round:

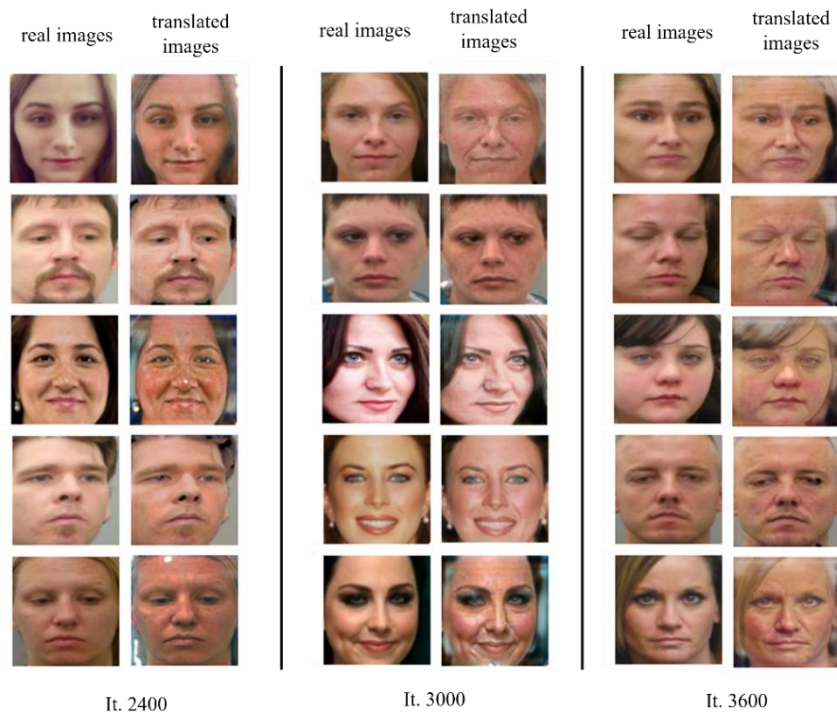


Figure 40 Results during the second stage of training of the GAN model at iterations 2400, 3000, 3600

Experiments conclusion

During the training, The results kept improving and the aging effects started to improve, in the early stages of training the colors started to change, it became less bright and the skin started to look a bit dull. As the training progressed, more aging started to show, like the hair color turned grey, the faces started to have wrinkles, dark spots, and hyperpigmentation.

5.2 Face Recognition

5.2.1 Face Recognition without aging:

Our main goal with this experiment is to establish a baseline performance for face recognition using Support Vector Machine (SVM), which is a well established method for classification tasks and provides a solid benchmark against more complex models and CNN which has been widely tested in face recognition.

Dataset

We tested the models against two benchmark datasets, LFW and Olivetti_faces.:

LFW Dataset Description:

Attribute	Description
Number of Images	13,233
Number of subjects	5,749
Image Resolution	250x250
Image Format	RGB
Annotations	Each image is annotated with the name of the person pictured. Some images have additional metadata such as gender, age, and pose.
Purpose	Designed for studying the problem of unconstrained face recognition, allowing testing of algorithms on real-world images.
Source	
Variations	Includes variations in lighting, facial expression, pose, occlusion (e.g., sunglasses), and background.
Usage	Commonly used for training and evaluating face recognition systems, particularly in research.

Olivetti face Dataset Description:

Attribute	Description
Number of Images	400
Number of subjects	40
Images per Subject	10
Image Resolution	64x64
Image Formate	Grayscale
Annotations	Each image is labeled with the subject ID. No additional metadata is provided.
Purpose	Created for facial recognition research, providing a controlled dataset for testing algorithms.
Source	Taken at the AT&T Laboratories in Cambridge between April 1992 and April 1994.
Variations	Includes variations in lighting, facial expressions, and poses.
Usage	Widely used for benchmarking facial recognition algorithms and machine learning research.

Results

	Olivetti Faces	LFW
SVM	N/A	76%
SVM+PCA	<u>96%</u>	86%
CNN	N/A	68.5%

5.2.2 Age Invariant Face Recognition

We tested some ML and DL models against aging datasets. The machine learning models didn't perform well, 6% was the highest

accuracy achieved using ML models. Deep learning models performed a lot better than deep learning models. Here are the results of the experiments we performed:

ML experiments

FGNET	
SVM	3%
Random Forest	<u>6%</u>

DL experiments

	Datase t	Optimize r	Loss Function	Callbacks	Accurac y
CN N	Our dataset	Adam	Categorical Crossentrop y	• Early stopping • ReduceLROnPlate au	93.89%

Chapter 6

Conclusion

6.1 Conclusion

In this project, we explored the application of Generative Adversarial Networks (GANs) to address the challenge of face aging for enhancing face recognition and verification systems. By leveraging the capabilities of GANs, particularly the CycleGAN architecture, we augmented our dataset with a wide variety of aged face images, thereby enabling our face recognition model to better understand and predict the effects of aging. This augmentation was critical in improving the model's performance, making it robust against age-related variations in facial features.

The integration of this augmented data into our face recognition and verification system demonstrated significant improvements in accuracy and reliability. Our system was tested extensively to ensure it meets the stringent requirements necessary for high-security environments like airport check-ins. The deployment of the model within a user-friendly GUI, developed using Tkinter, ensures ease of use for end-users while maintaining high standards of security and performance.

The project showcased the effectiveness of combining advanced machine learning techniques with practical applications, resulting in a comprehensive solution for age-invariant face recognition and verification. The system's ability to process real-time image capture, handle various input formats, and provide reliable verification results highlights its potential for broader deployment in security-critical applications.

6.2 Future Work

Moving forward, we aim to enhance our GAN model architecture by incorporating wavelet transformations and identity preservation

techniques. This approach is expected to further improve the quality and diversity of the augmented face images, providing even greater robustness to our face recognition and verification models. By leveraging wavelet transformations, we can capture more detailed features and subtle variations in the facial images, leading to better age progression and regression results.

Additionally, we plan to expand the functionality of our system to support more destinations, including those requiring printed visas. This enhancement will involve integrating advanced document scanning and verification technologies to accommodate international trips, thus broadening the system's applicability and utility. By addressing these future objectives, we aim to create a more versatile and powerful face recognition and verification system capable of meeting the evolving needs of security and identity verification in various contexts.

References

- [1] A. O. ,. A. B. a. A. T.-A. Insaf Adjabi, *Past, Present, and Future of Face Recognition*, pp. 3-4, 2020.
- [2] N. N. Praveen Kumar Chandaliya, *AW-GAN: face aging and rejuvenation using attention with wavelet GAN*, 9 August 2022.
- [3] M. V. Canal, *Using GANs for Face Aging*, pp. 14-15, January 19, 2020.
- [4] I. P.-A. J. M. M. X. B. W.-F. D. O. S. .. & B. Y. 1. Goodfellow, "Generative Adversarial Networks," 2014.
- [5] M. B. M. Albert Swiecicki, "A generative adversarial network-based abnormality detection using only normal images for model training with application to digital breast tomosynthesis," May 2021.
- [6] A. Sharma, "Conditional GAN (cGAN) in PyTorch and TensorFlow," JULY 12, 2021 .
- [7] L. M. S. C. Alec Radford, "Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks," 2016.
- [8] J.-Y. Z. T. Z. A. A. E. Phillip Isola, "Image-to-Image Translation with Conditional Adversarial Networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

- [9] J.-Y. a. P. T. a. I. P. a. E. A. A. Zhu, Unpaired image-to-image translation using cycle-consistent adversarial networks, 2017, pp. 2223-2232.
- [10] A. Upreti, "Convolutional Neural Network (CNN). A Comprehensive Overview," August 2022.
- [11] S. Z. Yanmei Li, Applied Research Methods in Urban and Regional Planning.
- [12] I. F. R. Antonino Guerrisi, "Artificial Intelligence and Advanced Melanoma: Treatment Management Implications," December 2022.
- [13] J. H. E. Simon J D Prince, *TOWARDS FACE RECOGNITION AT A DISTANCE*, July 2006.
- [14] D. H. Y. W. A. K. J. Hongyu Yang, *Learning Face Age Progression: A Pyramid Architecture of GANs*, Oct 2018.
- [15] L. M. L. L. Chenggang Yan, *Age-Invariant Face Recognition by Multi-Feature Fusion and Decomposition with Self-attention*, 25 January 2022.
- [16] J. R. Quinlan, "Induction of decision trees. Machine Learning,," 1986.
- [17] B. S.-R. Schölkopf, "Kernel Principal Component Analysis.,," 1997.
- [18] O. a. A. G. Daniel G. Costa, "A Discrete Wavelet Transform (DWT)-Based Energy-Efficient Selective

Retransmission Mechanism for Wireless Image Sensor Networks," February 2012.

- [19] T. P. P. I. A. A. E. J. Y. Zhu, "Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [20] I. P.-A. J. M. M. X. B. W.-F. D. O. S. C. A. & B. Y. Goodfellow, "Generative adversarial nets," *arXiv preprint arXiv:1406.2661*, 2014.
- [21] P. a. Z. J.-Y. a. Z. T. a. E. A. A. Isola, "Image-to-image translation with conditional adversarial networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 1125--1134.
- [22] A. K. A. I. A. A. Z. RASHA RAGHEB ATALLAH, *Face Recognition and Age Estimation Implications of Changes in Facial Features*, p. 6, June 19, 2018.
- [23] E. Newton, "Why Is TSA Adding Facial Recognition at Airports?," 17 3 2023. [Online]. Available: <https://www.airwaysmag.com/legacy-posts/tsa-facial-recognition-airports>.
- [24] M. B. J.-L. D. Grigory Antipov, *Face Aging With Conditional Generative Adversarial Networks*, 30 May 2017 .
- [25] G. Menghani, *Efficient Deep Learning: A Survey on Making Deep Learning Models Smaller, Faster, and Better*, 21 Jun 2021.

- [26] A. G. a. M. M. Toshnall Meenpal, "Spatial Domain Representation for Face Recognition," September 2019.
- [27] D. P. Bentley, "How do machine learning GANs work?," January 11, 2023. [Online]. Available: <https://www.sciencefocus.com/future-technology/how-do-machine-learning-gans-work>.
- [28] A. Vidhya, "A Review of Generative Adversarial Networks," 12 2020. [Online]. Available: <https://medium.com/analytics-vidhya/a-review-of-generative-adversarial-networks-part-1-a3e5757a3dc2>.
- [29] B. Hansen, "Cycle GAN with PyTorch," 2020. [Online]. Available: <https://towardsdatascience.com/cycle-gan-with-pytorch-ebe5db947a99>.